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Article title: Predicting Financial Distress and Bankruptcy in Small and Medium Enterprises: Insights from Machine Learning Models

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Predicting Financial Distress and Bankruptcy in Small and Medium Enterprises: Insights from Machine Learning Models

ABSTRACT

This study uses advanced machine learning techniques to predict financial distress and bankruptcy among small and medium enterprises. Leveraging a dataset spanning 2013–2020, we integrate accounting-based predictors, macroeconomic variables, and industry and geographic controls to enhance predictive accuracy. A comparative analysis of traditional statistical models and ensemble learning methods such as Random Forest, Adaptive Boosting, and Extreme Gradient Boosting demonstrates that machine learning models can predict financial distress and bankruptcy up to three years in advance. Our findings reveal that while bankruptcy prediction remains relatively stable over time, financial distress forecasting exhibits declining accuracy in the medium term, underscoring the complexity of early-stage distress identification. Feature importance analysis highlights the crucial role of macroeconomic indicators, reinforcing the necessity of incorporating broader economic conditions into predictive frameworks. This study contributes to the literature by distinguishing financial distress from bankruptcy using legally defined classifications to provide regulators, investors, and policymakers with practical insights into developing more effective early warning systems.

Keywords: machine learning, bankruptcy prediction, financial distress prediction, feature importance, macroeconomic indicators.

JEL classification: C53, G32, G33, G34.

1. Introduction

The ability of company owners, investors, regulators, and policymakers to predict corporate bankruptcy and financial distress is critical in financial economics. This paper investigates whether machine learning (ML) algorithms can accurately forecast bankruptcy and financial distress over short-term and medium-term horizons. We address two key research questions: (1) Can ML techniques outperform traditional statistical models in predicting corporate distress? (2) What are the key predictors of financial distress and bankruptcy in small and medium enterprises (SMEs)?

Despite the extensive and growing body of literature on firm failure prediction, several gaps remain. A primary difficulty is the absence of consensus on the definition of financial distress, with different regulatory regimes and academic studies using terms such as “bankruptcy,” “financial distress,” and “default” interchangeably (Balcaen and Ooghe 2006; Dichev 1998). Although research on financial distress prediction is steadily increasing, most studies focus on firms that exit the market through bankruptcy or liquidation, often using the term “financial distress event” interchangeably with these outcomes. However, while severe bankruptcy is clearly identifiable, mild forms of insolvency are much harder to detect and measure. This inconsistency across studies and regulatory frameworks complicates the development of reliable early warning systems that could help stakeholders mitigate financial distress before it escalates into bankruptcy.

Corporate failures can result in substantial economic and social losses because distress in one firm can propagate to financial institutions, supply chains, and the broader economy through contagion (Tian and Yu 2017). Minimizing these negative externalities requires reliable early warning systems that enable firms, financial institutions, and regulators to take timely corrective actions. However, existing prediction models often fail to distinguish between different levels of distress, limiting their practicality. This paper addresses these challenges by integrating advanced ML algorithms with a regulatory-aligned classification system, thus improving predictive accuracy and applicability.

The prediction of corporate failure has evolved significantly over the past five decades. Early studies applied multivariate discriminant analysis (MDA) (Beaver 1966; Altman 1968; Altman et al. 1977), logistic regression (Ohlson 1980; Zavgren 1985), and probabilistic regression (Zmijewski 1984). Over time, scholars incorporated hazard models to capture time-dependent bankruptcy probabilities (Campbell et al. 2008; Beaver et al. 2005; Shumway 2001).

More recently, ML algorithms such as Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Adaptive Boosting (AdaBoost) have gained popularity due to their ability to model complex, non-linear relationships without strong statistical assumptions (Alfaro et al. 2008; Gepp et al. 2010; Barboza et al. 2017). Despite these advances, logistic regression (also known as the logit model) and MDA are

still widely used in practice.¹ Simultaneously, an increasing number of predictive variables are augmenting these models. Early models began with exclusively financial ratios (Altman 1968; Ohlson 1980) as well as firm age and size (Modina and Pietrovito 2014), while later models introduced macroeconomic indicators (Hackbarth et al. 2006; Carling et al. 2007; Bruneau et al. 2012; Tinoco and Wilson 2013a; Carvalho et al. 2022), market-based indicators (Shumway 2001), and indicators related to corporate governance and innovation (Ciampi 2015; Bai and Tian 2020). More recently, Citterio and King (2023) have shown that ESG (Environmental, Social, and Governance) indicators improve the accuracy of financial distress prediction in banks, relying on the Z-score model rather than formal legal classifications such as bankruptcy filings or restructuring procedures.

Despite the vast amount of literature on ML models over recent years, scholars continue to create new classifiers or improve the existing models to predict bankruptcy more accurately. However, empirical evidence on the superiority of ML models remains inconclusive, given their performance varies by data structure, sample characteristics, and variable selection strategies (Duénez-Guzmán and Vose 2013).

This paper makes several contributions to the literature on financial distress and bankruptcy prediction. First, it advances the existing research by distinguishing between financial distress and bankruptcy based on regulatory classifications rather than arbitrary financial metrics. While prior studies often define firm failure using financial ratios such as return on assets (ROA) or leverage, this study aligns distress classification with legal procedures (or judicial actions) filed by firms, ensuring a more accurate and policy-relevant approach. By doing so, the paper enhances the applicability of predictive models to real-world financial decision-making, particularly for regulators, creditors, and policymakers seeking an early warning system that complies with national bankruptcy laws.

Second, this study extends the prediction horizon beyond the short term, which has always been the focus of bankruptcy prediction models. Traditional models are generally accurate for up to one year and, in some cases, up to two years (Altman et al. 2014; Shumway 2001; Nam et al. 2008). Some papers using ML techniques obtained good results in predicting firm failure in the short term (Barboza et al. 2017; Son et al. 2019). This paper investigates whether ML models can predict financial distress and bankruptcy with reasonable accuracy up to three years before the event. By incorporating a longer forecasting window, the findings contribute to more proactive risk management strategies that allow firms and financial institutions to implement timely corrective measures.

Third, this study employs an ML-based feature selection approach to identify the most influential predictors of financial distress and bankruptcy. Unlike traditional models, which rely on a predefined set of financial indicators to improve statistical accuracy and address dimensionality, this research leverages advanced ML techniques to analyze a high-dimensional dataset encompassing over eighty financial, macroeconomic, geographic, and industry-specific variables. It allows the ML algorithms to

¹ For a detailed literature survey, see Jones (2023).

determine independently the most relevant predictors, thus enhancing predictive accuracy and capturing the complex, non-linear relationships conventional models may overlook. Moreover, it mitigates the risk of prematurely excluding variables that could reveal arresting patterns in firm distress dynamics, offering a more data-driven and adaptable method for assessing risk. This approach is particularly relevant for SMEs as they are usually not listed firms, rendering market-based measures based on Black-Scholes-Merton models unsuitable.

Lastly, we tested our research questions on a sample of Italian SMEs from 2013 to 2020. This empirical application to Italian SMEs provides valuable insights into a segment of firms particularly vulnerable to financial distress. In fact, the probability of financial distress and bankruptcy is more common for SMEs than large companies, even the listed ones. Recent papers demonstrate that business groups can better manage intra-group credit risk (by reallocating resources within the corporate structure) than stand-alone firms such as SMEs (Beaver et al. 2024; Kumar 2024). Moreover, while SMEs represent a significant portion of the European economy, research on bankruptcy prediction in this sector remains limited. According to the Organisation for Economic Co-operation and Development (OECD (2020)), nine out of ten enterprises currently active in the European Union (EU) are SMEs, amounting to approximately 21 million enterprises. In the EU, Italy has one of the highest numbers of SMEs, constituting 99% of non-financial enterprises and producing 66.9% of Italian value-add compared to the 56.4% of the EU average. In addition, the share of Italian SME employment is higher than the EU average: 78.1% and 66.6%, respectively. Other papers, such as Becchetti and Sierra (2003), Barboza et al. (2017), and Perboli and Arabnezhad (2021), have adopted the Italian sample of SMEs to address similar research questions. Our paper takes a different, twofold perspective. First, it aims to identify an early warning system aligned with national bankruptcy regulation. Second, it tracks the juridical procedures undertaken by each firm. The findings have practical implications for investors, policymakers, and financial institutions because they highlight the potential of ML models to enhance credit risk assessment, improve financial stability, and refine regulatory frameworks for corporate distress management.

The remainder of the paper is structured as follows. Section 2 outlines the institutional framework of Italian bankruptcy regulation. Section 3 reviews the relevant literature and formulates the research hypotheses. Section 4 explains the methodology. Section 5 details the sample selection and data processing procedures. Section 6 presents the empirical results, and Section 7 the robustness checks. Section 8 concludes the paper.

2. Institutional setting

2.1 Financial procedures according to Italian business law

The Italian corporate failure framework, which has evolved through multiple reforms since 1942, primarily aims to improve creditor–debtor arrangements while reducing judicial intervention in corporate crisis resolutions. The most significant reforms—Law No. 80/2005 and Law No. 69/2013—amended the original Bankruptcy Law (Decree No. 267/1942) to promote out-of-court settlements, mitigating traditionally prolonged procedures with low recovery rates for creditors.

Italian law permits various contractual agreements to facilitate corporate restructuring, from debt reorganization to business continuity measures and asset liquidation (Campobasso 2017). The framework includes five main financial distress procedures: bankruptcy, administrative compulsory liquidation, judicial composition with creditors, debt restructuring agreements, and certified recovery plans. The primary rationale behind these reforms was to provide a diverse set of bankruptcy procedures that could accommodate different causes, severity levels, and potential resolutions of corporate distress (Campobasso 2017).

A more recent regulatory milestone was introduced in 2021 when Italy implemented a new bankruptcy and insolvency framework. This updated legislation emphasizes identifying the nature and causes of financial distress and classifying crises as either reversible or irreversible. Based on this distinction, these procedures can be categorized into restructuring and liquidation, as summarized in Figure 1.

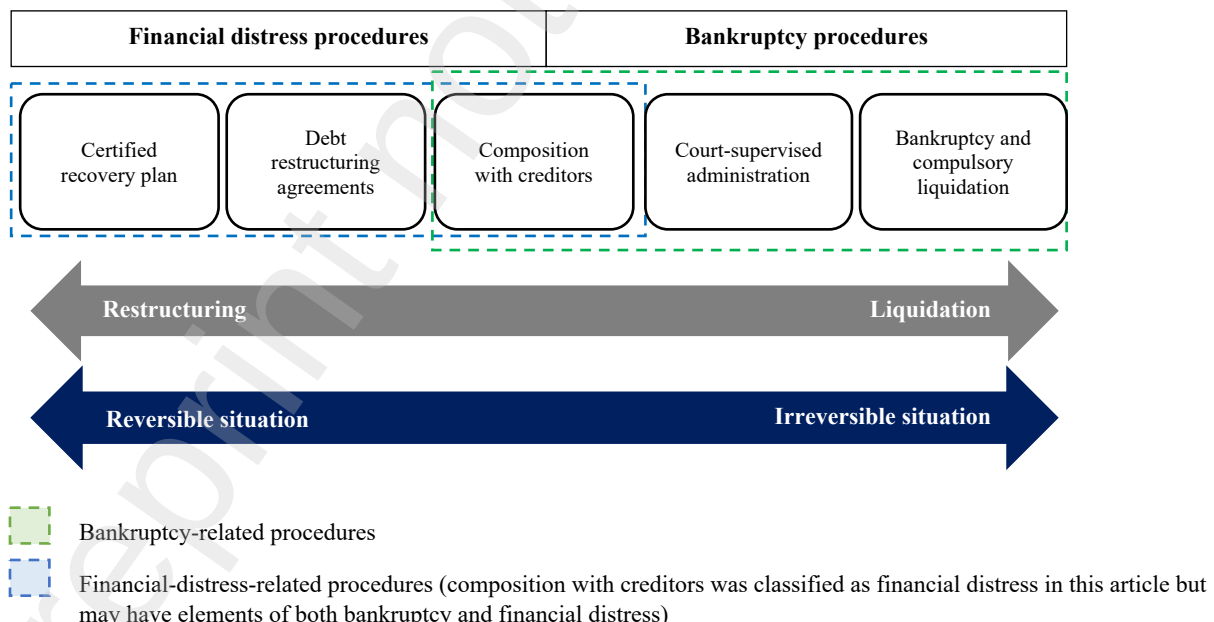


Fig. 1. Distress procedures according to Italian bankruptcy law.

The liquidation procedures include bankruptcy and administrative compulsory liquidation, which aim to remove financially unviable firms from the market by liquidating assets and distributing proceeds among creditors. They are court-based procedures. Bankruptcy is the most applied court-based procedure, while administrative compulsory liquidation is reserved for firms subject to government oversight. Both procedures typically apply to companies that can no longer ensure business continuity or whose financial situation is beyond recovery (Campobasso, 2017).

The remaining three procedures are plan-based restructuring procedures—certified recovery plans (CRPs), debt-restructuring agreements (DRAs), and judicial compositions with creditors—designed to reorganize the firm's financial position rather than liquidate its assets. Restructuring procedures vary in the degree of court involvement:

1. CRPs are informal, private agreements between creditors and debtors and do not require court approval unless voluntarily registered in the Business Register (*Registro delle Imprese*).
2. DRAs are semi-formal procedures involving judicial oversight to ensure legal compliance and requiring a petition to the court and approval from creditors representing at least 60% of total debt exposure.
3. Judicial compositions with creditors are formal procedures that entail full court supervision. An insolvency practitioner evaluates the plan, and the court ensures its feasibility. The plan may lead to either the continuation of operations through asset transfers or an orderly liquidation.

Semi-formal and informal procedures feature bounded court involvement, while formal restructuring processes require higher court intervention (Danovi et al. 2020).

According to the Italian system, an informal restructuring procedure may be conducted entirely through flexible contractual agreements between parties. An example of an informal restructuring procedure is a CRP. A debtor may draft a plan selecting the creditors involved and the operations in the CRP (e.g., injection of cash from new creditors or shareholders or sale of non-strategic assets). The CRP only binds the creditors who voluntarily approve the plan. In addition, the negotiations to establish a CRP are purely private, and the law does not require public disclosure, resulting in a fully confidential plan.

Over the past two decades, legislative changes—such as Law No. 134/2012—have refined restructuring mechanisms to encourage earlier interventions in financial distress and support business continuity.

In this paper, we classify firms based on the degree of intervention required by their legal procedures, distinguishing between i) bankrupt firms, namely firms subject to court-based liquidation procedures, and ii) financially distressed firms, namely firms subject to plan-based restructuring procedures with the exclusion of CRPs that are purely private and where the law does not require public disclosure. This

classification enables a sufficiently large sample for training ML models while preserving the distinct characteristics of court-driven versus contract-based resolutions.

2.2 Consequences of the COVID-19 outbreak for Italian firms

At the end of January 2020, Italy confirmed the first cases of COVID-19. The pandemic quickly spread across the northern regions before engulfing the entire country, although its effects varied significantly across local economies (OECD 2020). A March 2020 survey of small Italian businesses revealed that 72% of the 6,000 surveyed companies were severely affected, experiencing both a sharp decline in demand and severe disruptions along their supply chains. The government's response included progressively stricter containment measures, culminating in the March 2020 DPCM (Decree of the President of the Council of Ministers), which mandated the closure of all retail trade businesses except those providing essential goods. Additionally, industrial production was halted for all non-essential sectors, including those outside the utilities and agro-food industries.

COVID-19 profoundly disrupted both the supply and demand sides of the economy, with SMEs being particularly vulnerable due to their limited resilience to economic shocks and lack of flexibility in responding. Unlike larger firms, SMEs typically face obstacles in accessing capital, making them more susceptible to liquidity crises and insolvency. Recognizing these risks, the Italian government introduced two major financial support packages to mitigate the pandemic's economic fallout for SMEs (OECD 2020).

The first package, announced on March 17, 2020, included a moratorium on loan repayments and an expansion of the National Guarantee Fund for SMEs to enhance liquidity support. This initiative aimed to provide immediate financial relief by freezing loan repayments and extending credit guarantees. The second package, introduced in April 2020 under the so-called Liquidity Decree, sought to further ease firms' access to credit by leveraging state-backed guarantees through SACE, a public financial institution. These guarantees covered 70% to 90% of loan amounts, depending on company size and risk exposure (OECD, 2020).

One of the main financial weaknesses identified among Italian SMEs during this severe economic downturn was their chronic undercapitalization (Orlando and Rodano 2020). However, two studies (Giacomelli et al. 2021; Orlando and Rodano, 2022) indicate that the number of bankruptcies and liquidations in 2020 was surprisingly lower than in 2019, decreasing by 33% and 27%, respectively. The decline was particularly pronounced in March and June 2020, coinciding with the temporary suspension of court activities. In total, fewer than 24,000 bankruptcy proceedings in 2020 were initiated by SMEs, compared to approximately 31,000 in 2019. A more detailed examination of bankruptcy liquidation procedures revealed a reduction from 15,000 cases in 2019 to 13,000 in 2020 (Orlando and Rodano,

2020).² The downward trend in bankruptcy filings persisted into 2021, with total cases remaining below pre-pandemic levels. While this can be attributed partly to the temporary suspension of judicial proceedings during the height of the pandemic, government intervention probably played a significant role in preventing a wave of insolvencies. However, given the temporary nature of these measures, the long-term effects on firm survival remain uncertain. As financial support mechanisms phase out, it remains to be seen whether SMEs can sustain recovery or if latent financial distress will translate into increased bankruptcies in the coming years.

3. Literature review and development of hypotheses

This study applies a quantitative research design to investigate the predictors of corporate financial distress and bankruptcy. The methodological framework builds on two streams of the literature: (i) ML models for bankruptcy prediction and (ii) financial distress and bankruptcy determinants.³

3.1 Models used to predict a firm's bankruptcy based on AI

Research into predicting bankruptcy has evolved significantly, incorporating various modeling approaches over time. Early studies often relied on single, complex models, such as neural networks (NNs), to predict bankruptcy. These models can capture intricate, non-linear relationships in data but can be computationally intensive and difficult to interpret. More recently, the focus has shifted toward ensemble techniques such as decision trees (DTs), which combine multiple simpler models to improve prediction accuracy and reduce the risk of overfitting. DTs offer enhanced transparency but might struggle with certain forms of non-linearity unless combined or boosted. Bagging and boosting in ensemble models have shown promise in achieving superior performance by leveraging the strengths of the different classifiers. A key distinction across these three approaches—NNs, DTs, and ensembles—lies in balancing predictive accuracy against interpretability. Ensemble techniques typically provide best-in-class accuracy, but their internal mechanics can become more difficult to interpret than a simple decision tree—although feature importance in Random Forest and boosting algorithms can partially address this limitation. This section reviews the key methods employed in bankruptcy prediction, emphasizing both single-model approaches like NNs and the more advanced ensemble techniques that have gained prominence in recent research.

² The authors included bankruptcy filings in their analysis, even though these do not appear in the Italian Business Register because they occur before formal bankruptcy declarations. In contrast, bankruptcy declarations are official judicial statements that initiate the legal procedure and are, therefore, recorded in the Italian Business Register. To assess the economic impact of the pandemic, the authors constructed a counterfactual scenario using 2019 accounting data, simulating a baseline without external shocks.

³ In this study, we do not provide a detailed review of traditional bankruptcy prediction models because there are comprehensive and thorough reviews by Alam et al. (2021); Jones (2023). Also, Jones et al. (2015, 2017) comprehensively compare alternative ML methods, including describing each major class of model, hyper-parameter estimation features, and the model-fitting process for each method.

NNs and DTs belong to the first group. More specifically, NNs were among the earliest applications of AI (artificial intelligence) in predicting financial distress. Their layered architecture, consisting of interconnected nodes, enables them to capture highly non-linear relationships within financial data. A study by Tam and Kiang (1992) predicting Texas bank failures, using a sample of 59 healthy and 59 failed banks from 1985 to 1987, demonstrated promising results that spurred further research in the field. Subsequent studies, including those by Wilson and Sharda (1994) and Zhang et al. (1999), confirmed that NNs could outperform traditional techniques like discriminant analysis and logistic regression, particularly when trained on sufficiently large datasets. As stated above, NNs, despite their effectiveness, are often computationally intensive and opaque, which makes it challenging for practitioners to interpret how specific inputs influence the outcomes. In contrast, DTs provide a more interpretable approach, partitioning datasets into nodes based on thresholds of predictor variables. This creates a flowchart-like structure that makes the decision process more transparent. Studies such as Gepp et al. (2010) have shown that DTs outperform traditional models like MDA in terms of accuracy. Additionally, Li et al. (2010) validated the effectiveness of DTs—specifically, the classification and regression trees (CART) algorithm—for failure prediction by comparing it to logit, k -nearest neighbors (k -NN), and support vector machine (SVM) using a sample of 270 Chinese public companies, 135 of which had failed. The results show that CART outperformed MDA, logit, and k -NN, and marginally outperformed SVM. However, DTs can overfit the data if they become too complex, necessitating techniques like pruning to ensure generalizability.

Building on both NNs and DTs, more recent studies have advanced bankruptcy prediction using ensemble techniques. These generate multiple training datasets through feature sampling (*bagging approach*) or resampling mechanisms (*boosting approach*). They combine the unique advantages of individual classifiers while reducing error variance.

Common ensemble techniques include algorithms such as RF (a bagging approach), AdaBoost, and XGBoost (boosting approaches). Research consistently highlights the superior performance of ensemble methods in bankruptcy prediction. For instance, Alfaro et al. (2008) pioneered the application of AdaBoost for bankruptcy prediction in Spanish companies active between 2000 and 2003, 590 of which failed during this period. AdaBoost demonstrated a test error of 8.90%, outperforming NNs by reducing the error by 30%. In contrast, Zięba et al. (2016) compared a range of classifiers, including logit, SVM, DT, and NN, along with AdaBoost and XGBoost, to a dataset of Polish companies between 2000 and 2012. Both AdaBoost and XGBoost performed better than the other single classifiers, with XGBoost showing superior performance. In contrast to Zięba et al. (2016), studies by Jones et al. (2017) and Barboza et al. (2017) compared ensemble methods alongside single classifiers on U.S.-listed firm data. Both studies also included RF algorithms, a bagging technique that operates like a boosting algorithm. The principal intuition behind RF is that, in a bagging process with DTs, strong predictors are likely to

appear in most or all the trees in the top section, thus making the bagged trees more comparable and simplifying the process of selecting the most relevant predictors. Jones et al. (2017) concluded that ensemble classifiers such as AdaBoost, generalized boosting, and RF performed more sharply than all other classifiers. According to the achieved accuracy, sensitivity, and specificity of boosting across all the approaches, the standout model is generalized boosting (average AUC=0.931). AdaBoost ranks very near with an average AUC of 0.928. RF is ranked third (average AUC=0.919). Conversely, logit and linear discriminant analysis (LDA) rank as the worst-performing classifiers (Jones et al. 2017). Similarly, Barboza et al. (2017) conclude that ensemble techniques such as bagging, boosting, and RF have higher accuracy than other classifiers. All the above algorithms perform very well if the sample is balanced and without outliers—two relevant issues in the context of bankruptcy analysis.

More recently, Alam et al. (2021) proposed a deep learning model using a panel data structure to predict corporate failure from a sample of 641,667 firm-month observations of North American listed companies between 2001 and 2018. The authors demonstrated these methods could reach a higher level of accuracy than a discrete hazard model and can exploit time dependency in the dataset.

In practical settings, model choice ultimately depends on data availability, computational resources, and stakeholder needs for explainability. NNs remain well-suited to large-scale problems where intricate data interactions exist; DTs can be deployed quickly and understood easily; ensemble methods strive to balance accuracy and robustness. Together, these approaches represent the evolving frontier of financial distress prediction, offering varied methodological options to researchers and practitioners alike.

3.2 A firm's financial distress and bankruptcy determinants

Using financial ratios to predict bankruptcy has a long history (Beaver 1966). Empirical studies emphasize the predictive relevance of financial ratios, categorized into profitability, solvency, liquidity, and efficiency (Beaver 1966; Altman 1968; Wilcox 1971; Deakin 1972; Altman and Narayanan 1997), along with other firm characteristics such as age and size (Modina and Pietrovito 2014). Beaver et al. (2012) contribute to this literature by considering additional factors such as differences in discretion, unrecognized intangible assets, book-to-market ratios, and incidence of losses on the predictive ability of financial ratios for bankruptcy. They find that discretion in financial reporting is associated with a significant deterioration in the predictive power of the accounting-based model. More recently, macroeconomic variables have been recognized as relevant predictors of corporate failure (Carvalho et al. 2022; Bruneau et al. 2012; Tinoco and Wilson 2013b).

3.2.1 Accounting variables

Regarding profitability, Altman et al. (1977) were among the first to investigate the discriminant power of the earnings before interest and tax (EBIT) to total assets ratio. Subsequent empirical research has expanded on this foundation by examining additional profitability indicators, including net income

to total assets, also known as return on assets or ROA (Ohlson 1980), operating income to total assets, also known as operating return on assets or OROA (Theodossiou et al. 1996), retained earnings to total assets (Altman 1968; Altman et al. 1977), and profit to total assets (Deakin, 1972; Beaver, 1966). These ratios collectively provide a comprehensive view of a firm's ability to generate returns from its resources and serve as key predictors of financial health or distress.

Solvency is related to the firm's ability to repay its long-term debt. A firm's solvency concerns the total amount of debt issued, the capital structure, and the earning adequacy to cover the expenses related to the issued debt (e.g., interest and various fixed charges such as lease payments). Leverage can amplify the effect of having fixed costs and is classified as either operating or financial leverage. The first derives from fixed costs in the firm's business operations. If sales increase and the firm has operating leverage, operational income increases exponentially due to stationary fixed costs. Vice versa, financial leverage concerns the inclusion of debt in the capital structure to decrease the firm's cost of capital, resulting in higher returns to equity holders. According to Modigliani and Miller (1958), firms desiring to increase their leverage should ponder the probability of a default surge. Likewise, lenders will inflate borrowing costs to compensate for the greater credit risk. Specifically, the corporate failure probability surges when the return on debt (ROD) is deficient and considerably lower than the return on investment (ROI). Deakin (1972) included leverage to predict financial distress, calculated as liabilities compared to equity and total assets. The author expected highly leveraged firms to become financially distressed. Another ratio included in this category is the coverage ratio. Coverage ratios determine the company's ability to meet interest payments on debt.

Liquidity focuses on the company's ability to satisfy its short-term obligations. Liquidity ratios measure how fast the firm can transform assets into cash. Low liquidity impairs the firm's ability to obtain credit and may interfere with normal business operations (Theodossiou et al. 1996). Overall, the required liquidity level depends on the type of industry and the firm's need for funds at any given time. In normal circumstances, the cash outflow is modest; however, in a market crisis, the cash discharge may increase considerably due to higher contingent liabilities related to the business bankruptcies and default proliferation that often characterize these events. Funding shortfalls aggravate crises, while liquidity levels tend to plunge. The liquidity ratios commonly used are the current ratio (Altman et al. 1977; Barnes 1990), the quick ratio (Barnes 1990), the cash ratio, and the defensive interval ratio. The models devised by Blum (1974), Beaver (1966), and (Deakin 1972) include supplemental ratios such as EBITDA (earnings before interest, taxes, depreciation, and amortization) coverage and cash flow over current liabilities.

Efficiency ratios are variables intended to assess how efficiently a firm manages its assets (i.e., its operational performance). The ratios measure the management of assets and working capital. Inefficiently managed and defaulting firms are common acquisition targets for acquirers with the know-

how to improve efficiency. Overall, inefficient firms are also expected to be more unprofitable than comparable efficient firms—consequently, more likely to be insolvent (Theodossiou et al. 1996). Usually, activity ratios have a direct impact on the assessment of liquidity. Efficiency ratios, such as turnover ratios, are calculated as a line item of the income statement divided by a line item of the balance sheet (Modina and Pietrovito 2014), such as inventory to sales (Theodossiou 1993). Other measures of efficiency compare added value and the number of employees.

3.2.2 Macroeconomic variables

Economic intuition may suggest that macroeconomic factors play a relatively significant role since the business cycle should greatly influence a firm's default risk. Several studies, over the years, have analyzed the effect of macroeconomic factors such as gross domestic product (GDP) growth, the output gap, and interest rates on the probability of corporate failure (Koopman and Lucas 2005; Hackbarth et al. 2006; Carling et al. 2007; Bruneau et al. 2012; Tinoco and Wilson 2013a; Carvalho et al. 2022).

Initially, Koopman and Lucas (2005) analyzed the dynamic relationship between the real economy and credit risk, modeling business failure probabilities alongside macroeconomic data. They identified distinct short-, medium-, and long-term credit risk cycles. Their findings revealed a strong co-cyclical pattern every six years between corporate defaults and interest rate spreads (Baa-rated corporate bonds versus bonds with maturities over ten years) and a similar correlation over longer cycles between GDP and default rates. Hackbarth et al. (2006) argued that macroeconomic factors significantly impact not only credit risk but also corporate financing decisions, with the optimal debt level depending on macroeconomic conditions. Carling et al. (2007) developed a duration model incorporating macroeconomic and firm-specific variables to estimate a firm's survival time before loan default. Using data from over 50,000 firms, they identified three key macroeconomic factors. First, the output gap, as an indicator of aggregate demand, showed that higher demand reduces default risk. Second, a positively sloped yield curve signaled favorable economic prospects, while a negative slope suggested economic slowdown, with widening short- and long-term interest rate spreads linked to lower default rates. Third, positive household expectations were associated with reduced corporate defaults because improved consumer confidence boosts spending. Their findings confirmed that macroeconomic indicators enhance default risk prediction beyond traditional financial ratios, improving model accuracy by capturing absolute risk levels rather than merely ranking firms by default probability.

Bruneau et al. (2012) applied a logit model, similar to Shumway (2001), to analyze the relationship between bankruptcies in France and macroeconomic variables. Their model incorporated long-term interest rates, the output gap, the dollar/euro exchange rate, and inflation. Their findings showed that all macroeconomic indicators significantly influenced bankruptcy rates: a lower output gap led to a sharp rise in bankruptcies, and higher interest rates increased borrowing costs for firms, exacerbating financial distress.

Similarly, Tinoco and Wilson (2013a) developed a financial distress and bankruptcy prediction model for publicly traded U.K. firms, incorporating market variables, macroeconomic indicators, and financial ratios. Unlike Bruneau et al. (2012), they selected only two macroeconomic indicators: the short-term Treasury bill rate deflated and the retail price index (RPI). While the RPI represents inflation, its relationship with bankruptcy rates remains ambiguous. Some economists argue that inflation encourages investment, reducing bankruptcy risk, because investors prefer to invest rather than lose purchasing power. Others contend that inflation signals weak macroeconomic conditions, increasing default probability (Mare 2015). Carvalho et al. (2022) capture the informational value of macroeconomic factors on credit default prediction by using data from firms spanning 11 European countries and show that macroeconomic information strengthens the accuracy of models forecasting the credit default of non-financial firms. With a negative effect on the probability of default, GDP growth stands out among the key macroeconomic predictors of default.

3.3 Hypotheses development

Building on existing research in ML applications for financial distress prediction, this study formulates three hypotheses to evaluate the predictive performance of ML models and to explore the distinct characteristics of firms experiencing financial distress or bankruptcy. The first hypothesis addresses the predictive capabilities of ML methods related to conventional statistical techniques. Traditional approaches frequently assess corporate failure risk only up to 12 months before a bankruptcy filing, relying mainly on data from the year immediately preceding the judicial procedure. By contrast, evidence indicates that ML algorithms can capture complex interactions among firm-specific and macroeconomic variables, often resulting in higher accuracy (Alaka et al. 2018; Barboza et al., 2017). Furthermore, the ability of newer ML methods to incorporate richer datasets and advanced algorithms suggests a potential extension of the predictive horizon beyond one year. Consequently, we hypothesize:

H1: Machine learning models can predict financial distress and bankruptcy with a reasonable level of accuracy in the short and medium term, up to three years before the judicial procedure.

The second hypothesis investigates whether the predictors of financial distress diverge from those driving bankruptcy. Although the two phenomena are interrelated, the literature indicates that distressed firms may suffer from acute liquidity constraints and high leverage, leading to restructuring, whereas bankruptcy is more frequently associated with solvency erosion and profitability decline (Bruneau et al. 2012; Tinoco and Wilson 2013a). In the context of the Italian legal framework—where firms can undergo distinct legal pathways (restructuring or liquidation)—we hypothesize:

H2: The financial and macroeconomic predictors of bankruptcy and financial distress differ, reflecting the distinct processes of restructuring (driven by short-

term liquidity shortages and elevated leverage) and liquidation (characterized by more severe solvency issues and profitability deterioration).

Additionally, this study incorporates the COVID-19 crisis as a natural experiment to assess the robustness of ML models in periods of extreme financial turbulence. The pandemic-induced economic downturn, coupled with policy responses such as loan moratoria and financial aid programs, significantly altered firm-level financial conditions (Perboli and Arabnezhad 2021). This raises the question of whether ML models trained on pre-COVID data retain their predictive accuracy in the post-pandemic period. Therefore, we formulate the following hypothesis:

H3: The financial distress caused by the COVID-19 outbreak has significantly impacted the predictive ability of machine learning models, potentially reducing their accuracy due to shifts in firm behavior and economic conditions.

By examining these hypotheses, this study aims to enrich the existing literature on ML-driven corporate failure prediction. The findings will shed light on the adaptability of ML approaches under varying economic climates and enhance our understanding of the distinct financial signals exhibited by firms approaching either restructuring or bankruptcy. The institutional complexity of the Italian setting makes it an ideal laboratory, offering insights extendable to various other contexts.

4. Methodology

This study employs a structured approach to applying ML models for predicting financial distress and bankruptcy, as illustrated in Figure 2. We began with data collection and preprocessing, proceeding to model implementation, performance evaluation, and interpretability assessment. The dataset comprises approximately 136,000 non-financial Italian firms active between 2013 and 2020. We combined accounting variables sourced from AIDA (*Analisi Informatizzata delle Aziende*), macroeconomic indicators from the OECD, and industry and geographic controls to capture sectoral and regional variations. To mitigate the impact of extreme values in financial ratios, we applied the hyperbolic tangent transformation, also known as the tanh function, following the approach proposed by Tinoco and Wilson (2013b). Missing values were imputed using the XGBoost-based approach, a computationally efficient ML-based approach to ensure data completeness and integrity.

Because financially distressed and bankrupt firms constitute less than 1% of the sample, we employed the synthetic minority over-sampling technique (SMOTE) to address the class imbalance (Chawla et al. 2002). This method generated synthetic observations of the minority class, thus balancing our dataset and reducing model bias toward the majority class. Additionally, we removed features exhibiting a correlation coefficient above ± 0.8 , limiting multicollinearity and thus preventing redundant predictors from skewing the training process.

We then implemented a range of ML classifiers to evaluate predictive performance, including ensemble learners such as AdaBoost, XGBoost, and Random Forest. For comparison, we tested more traditional classification techniques—logistic regression (both full and stepwise), k -NN, decision trees, and artificial neural networks. Each model was trained on 70% of the dataset and validated on the remaining 30% to ensure a robust, unbiased performance evaluation.

Model accuracy was measured using several metrics. The primary indicator was the AUC (area under the curve) of the receiver operating characteristic curve (ROC), which is particularly suited to imbalanced datasets. Additional evaluations included precision, recall, and the F1-score for classification accuracy, along with overall accuracy as a broader measure of model effectiveness. To assess predictive strength across different time horizons, we tested model performance one, two, and three years before the filing event ($t-1$, $t-2$, $t-3$).

Finally, the study incorporated both global and local model interpretability methods. Permutation feature importance (PFI) analysis identified the most influential predictors in the overall dataset (global interpretability), while the SHapley Additive exPlanations (SHAP) technique revealed how individual features drive specific classification outcomes (local interpretability). By adopting this comprehensive methodological framework, our study ensured its results were robust and transparent, in alignment with best practices for ML-driven financial distress and bankruptcy prediction.

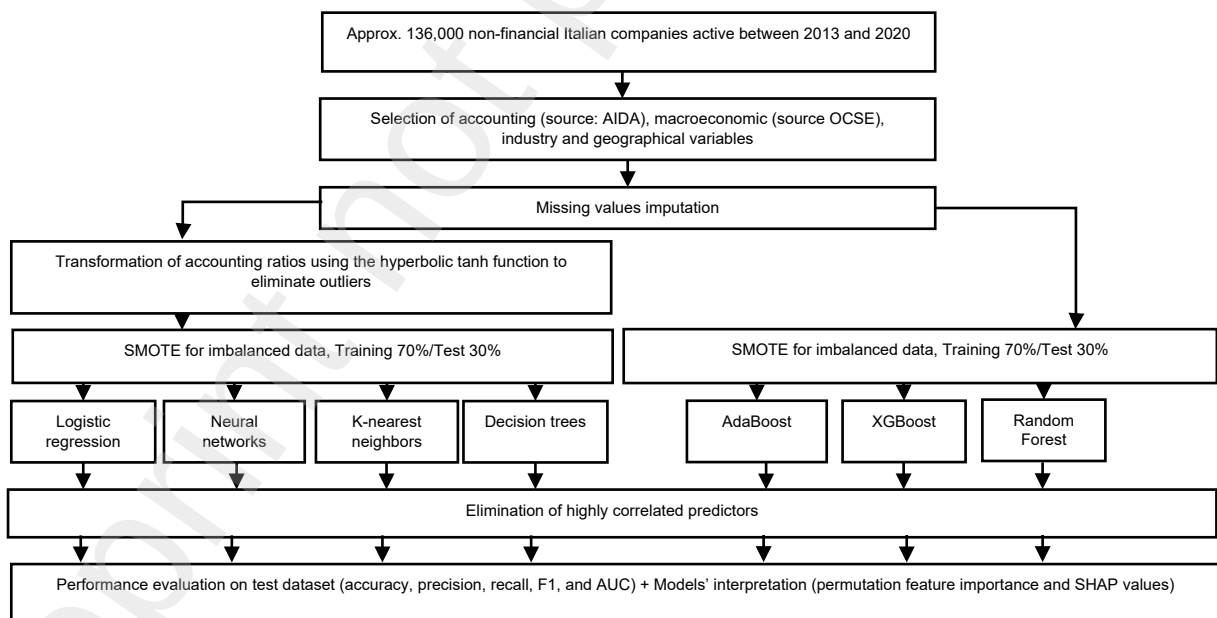


Fig. 2. Description of implemented algorithms.

5. Data collection

5.1 Sample selection

Our sample, drawn from Bureau Van Dijk's AIDA, comprises 130,000 non-financial Italian firms active from 2013 to 2020, corresponding to 998,730 firm-year observations. The selection criteria were: i) companies must have at least one available financial statement between 2013 and 2020; ii) companies should not be in the financial sector; iii) companies should have reached the following dimensional standard in at least one year between 2013 and 2020: a) €2 million in total revenues and b) €2 million in total assets. Because financial data are considered fundamental for estimating the effects and determinants of bankruptcy, the sample includes only companies with available financial statements. Moreover, the dimensional standard (i.e., criterion iii) reflects EU Recommendation 2003/361 concerning the definitions of microenterprise, small enterprise, and medium-sized enterprise.⁴

The AIDA database also contains information on the type of judicial events filed by unhealthy companies. In other words, we know the financial distress procedure filed for each company (i.e., bankruptcy, juridical composition with creditors, debt restructuring agreements, administrative compulsory liquidation, inability to repay creditors, or court-supervised administration) between January 1st, 2012, and December 31st, 2021.

Table 1 displays the distribution of healthy and unhealthy firms by geographic region and industry sector. Firms are classified as healthy or unhealthy based on their financial status. The classification is determined by where each firm is registered, reflecting its primary area of operations. Panel A shows that most firms (36.95%) are located in Italy's Northwest, while 25.30% are in the Northeast, 19.49% in the Center, and 18.26% in the South and Islands. Out of 135,848 firms, 4,901 (3.74%) are deemed unhealthy. Across regions, the proportion of unhealthy firms ranges from 3.27% in the Northwest to 4.37% in the Center, indicating that the Center exhibits the highest share of distressed or failing businesses relative to other areas. Panel B reports the distribution of firms based on the ATECO code, which is the statistical classification of economic activities defined by the Italian Institute of Statistics (ISTAT). Most firms operate in the manufacturing sector (32.65%), followed by wholesale and retail trade (24.78%). The third most common category is services (20.17%). The sample encompasses firms from all industries except the banking and financial sectors.

⁴ As microenterprises have modest figures, any changes to these figures could have a significant effect on financial ratios, which may become ineffective when assessing companies under a certain dimensional threshold (Ciampi 2015; Ciampi and Gordini 2013).

Table 1**Sample distribution by health status and geographic region.**

This table presents the distribution of healthy and unhealthy firms across Italy's four major geographic regions (Northwest, Northeast, Center, and South-Islands) and various industry sectors. Panel A disaggregates the sample by geographic region; Panel B classifies firms by industry. Column (I) shows the total number of healthy firms, and column (II) indicates the percentage of those healthy firms relative to all healthy firms in the sample. Column (III) reports the number of unhealthy firms, and column (IV) represents the percentage of those unhealthy firms relative to all unhealthy firms in the sample. Column (V) provides the total number of firms, and column (VI) shows the corresponding percentage of the overall sample. Finally, column (VII) presents the percentage of unhealthy firms within each region or industry.

	No. healthy firms	% healthy firms on total healthy firms	No. unhealthy firms	% unhealthy firms on total unhealthy firms	Total firms	% firms by geographic area	% unhealthy firms by geographic area
	(I)	(II)	(III)	(IV)	(V)	(VI)	(VII)
<i>Panel A</i>							
Northwest	48,556	37.08	1,644	33.54	50,200	36.95	3.27
Northeast	33,150	25.32	1,217	24.83	34,367	25.30	3.54
Center	25,324	19.34	1,157	23.61	26,481	19.49	4.37
South-Islands	23,917	18.26	883	18.02	24,800	18.26	3.56
Total	130,947	100	4,901	100	135,848	100	3.74
<i>Panel B</i>							
Agriculture	2,728	2.08%	92	1.88%	2,820	2.08%	3.37%
Construction	16,090	12.29%	979	19.98%	17,069	12.56%	6.08%
Manufacturing	42,853	32.73%	1,554	31.71%	44,407	32.69%	3.63%
Services Sector	26,529	20.26%	881	17.98%	27,410	20.18%	3.32%
Wholesale & Retail Trade	32,606	24.90%	1,053	21.49%	33,659	24.78%	3.23%
Transportation	6,612	5.05%	237	4.84%	6,849	5.04%	3.58%
Utilities	3,529	2.69%	105	2.14%	3,634	2.68%	2.98%
Total	130,947	100%	4,901	100%	135,848	100%	3.74%

Moreover, we categorize companies into three groups: i) healthy firms, ii) bankrupt firms, and iii) financially distressed firms. Firms subject to bankruptcy, administrative compulsory liquidation, or court-supervised administration are categorized as bankrupt. Firms subject to judicial composition with creditors, debt restructuring agreements, or inability to repay credits are categorized as financially distressed. Debt restructuring agreements and juridical compositions with creditors tend to pursue the firm's going concern. Bankruptcy, administrative compulsory liquidation, and court-supervised administration are usually concerned with liquidating the firm's assets. Table 2 shows the evolution of the judicial proceedings between 2013 and 2020.

Table 2.**Evolution of judicial proceedings among unhealthy firms (2013–2020).**

This table reports the yearly number of firms undergoing different legal proceedings—bankruptcy, administrative compulsory liquidation, court-supervised administration, judicial composition with creditors, debt restructuring agreements, and other distress-related events—from 2013 to 2020. The final rows present the total counts of unhealthy and healthy firms, as well as the percentage of unhealthy firms in the overall sample each year.

Year	2013	2014	2015	2016	2017	2018	2019	2020
	(I)	(II)	(III)	(VI)	(V)	(VI)	(VII)	(VIII)
Bankruptcy	12	43	61	53	250	231	258	166
Administrative compulsory liquidation	14	41	62	68	50	56	50	36
Court-supervised administration	13	23	23	11	14	8	5	7
Juridical composition with creditors	529	554	510	301	237	212	221	229
Debt restructuring agreements	74	72	62	62	70	71	78	63
Inability to repay creditors	9	18	33	41	56	48	41	25
Other financial distress-related events	0	0	0	0	3	2	1	4
# unhealthy firms	651	751	751	550	682	628	654	530
# healthy firms	126,557	125,456	125,205	123,697	121,469	118,779	115,583	111,990
% unhealthy firms	0.51%	0.60%	0.60%	0.44%	0.56%	0.53%	0.56%	0.47%

In line with prior literature, our database is imbalanced, with the share of unhealthy firms ranging from 0.44% (in 2016) to a peak of 0.60% (in 2014 and 2015). Over the past eight years, the absolute number of bankruptcy declarations has gradually risen, while the proportion of healthy firms has shown only a slight decrease.

5.2 Data processing

Data processing and quality control are crucial for obtaining valid and reliable results. We examined the collected data thoroughly to enhance the overall quality of the dataset. A validation analysis of bankrupt companies eliminated duplicates across samples, and a review of financial distress events removed additional redundancies. As a result, three key issues were addressed: missing values, extreme values, and data balancing.

5.2.1 Missing values

A key concern with the database is that some yearly observations are partially or entirely missing, which can occur for various reasons. For instance, a company may be liquidated and not file financial statements for all the years after, or it may simply omit filing one year's report. If all financial statement items for a particular year were missing, we removed that entire observation from the dataset because no meaningful inference was possible. This procedure eliminated 92,318 observations, reducing the

sample from 1,091,048 to 998,730 observations. Where only some data points were missing, we observed that unhealthy firms tended to have higher rates of missing data than healthy firms, as shown in Table 3. However, as removing them could undermine the model's ability to detect financial distress or bankruptcy patterns, we employed data imputation methods to fill in the missing values. Specifically, we used an XGBoost-based approach, which efficiently leverages gradient-boosted decision trees to capture potential interactions and nonlinear relationships (Deng and Lumley 2024).

Table 3

Missing values of unhealthy and healthy firms.

This table presents the number and percentage of observations with missing data for unhealthy (financially distressed or bankrupt) versus healthy firms.

	Unhealthy firms	Healthy firms
# observations with missing values	219,592	3,260,045
# total observations	3,359,328	108,498,432
% observations with missing values	6.54%	3.01%

5.2.2 Extreme values

Databases that aggregate financial statement data often contain extremely high or low values. These outliers can lead to unusually large residuals in logistic regressions and may affect the accuracy of maximum likelihood estimations, resulting in invalid inferences. To address this, we applied the hyperbolic tangent function, $\tanh(x)$, to the dataset. This transformation is particularly effective in mitigating the influence of extreme values (Tinoco et al. 2018). However, algorithms such as XGBoost, AdaBoost, and Random Forest are inherently robust in handling outliers and noisy data. Therefore, for these models, we relied on the original (unscaled and untransformed) dataset to train and test our models.

5.2.3 Data balancing

The last concern regards data balancing. Our dataset is highly imbalanced, with bankrupt or financially distressed companies constituting less than 1% of all observations. ML algorithms often struggle with imbalanced datasets, as they fail to glean meaningful patterns for the minority class and tend to favor the majority class (Abedin et al. 2021). To address this issue, we applied SMOTE, introduced by Chawla et al. (2002). SMOTE creates synthetic samples by interpolating between randomly selected k -NN within the minority class based on the desired oversampling rate. We then combined this approach with under-sampling of the majority class to reduce classifier bias further and improve predictive performance for minority observations.

5.3 Selected variables

Our analysis relies on three categories of explanatory variables: (i) financial ratios, (ii) qualitative factors, and (iii) macroeconomic indicators. First, we collected 80 financial ratios (R1–R81) spanning five dimensions of firm performance—profitability, efficiency, solvency, liquidity, and output. After removing highly correlated measures, 61 ratios remained. Second, we included 11 qualitative variables capturing the firm’s industry sector and geographic macro-region. Finally, we identified 10 macroeconomic variables but retained only 5 in the final dataset to mitigate multicollinearity.⁵

6. Empirical analysis

6.1 Predictive performance in the short and medium term (H1)

Tables 4 and 5 summarize the performance measures. All the measures are calculated on the test dataset, corresponding to 30% of the total number of observations in the database, roughly 300,000. Since the test dataset is highly imbalanced and the minority class is approximately 1% of total observations, the most efficient method to evaluate classifier performance (see Section 3) is the area under the curve (AUC).⁶ We discuss our results based on the AUC score; however, for the sake of completeness, we report also the typical metrics used in the literature, namely accuracy, precision, recall, and F1-scores.

For the financial distress prediction, the ML classifiers—specifically, XGBoost, Random Forest, and AdaBoost—outperformed other models, such as k -NN, decision trees, and neural networks, across all time horizons.⁷ This is evident from their consistently high AUC scores, particularly at time $t-1$ (average score of 0.988), $t-2$ (average score of 0.947), and $t-3$ (average score of 0.891). Notably, the logit models (both full and stepwise) show similar performance, with AUC scores closely matching those of the top-performing ML classifiers. This finding is consistent with previous studies indicating that, despite their relative simplicity, logit models can provide competitive predictive power in the context of financial distress (Jones et al. (2017)).

It is worth highlighting two important aspects. First, AUC scores decrease over time, as expected when predicting financial events that occur in the future. This trend is more pronounced in financial distress prediction at $t-3$, where the AUC scores drop by approximately 10%. The decline may be attributable to the increasing difficulty of predicting financial distress further in advance, as companies in distress may show fluctuating financial health over the medium to long term. At $t-2$, the drop in AUC scores is less pronounced, around 4%. Likely contributing factors are data sparsity and the models’

⁵ The complete list of variables is provided in Table 1A of the Appendix A, where the variables excluded from the models are marked with a (*). Table 2A presents the descriptive statistics for financially distressed firms and bankrupt firms at time $t-1$, $t-2$, and $t-3$, alongside those for healthy firms.

⁶ ROC curves are shown in Appendix B, Figure 1B.

⁷ Except at $t-3$ where the neural network is the best performer.

reduced ability to capture the evolving dynamics of firms' financial conditions over time. The second important aspect is the highly imbalanced nature of the dataset used for training the models, with distressed firms constituting only about 1% of total observations. This imbalance means that the high AUC scores, especially for the ML classifiers, reflect the models' ability to distinguish distressed firms from healthy ones. However, this comes with precision-recall tradeoffs, particularly in the longer-term prediction at $t-3$. As observed, the models maintain a reasonable precision in the short term, but the recall measure decreases, suggesting that many distressed firms remain undetected in more distant predictions.

The bankruptcy prediction results mirror the findings for financial distress in many respects. The ML classifiers, particularly XGBoost, Random Forest, and AdaBoost, perform similarly to logit models in terms of AUC, though slightly lower than for financial distress prediction. At $t-1$ and $t-2$, the AUC scores for bankruptcy prediction are lower than for financial distress, probably due to the more sudden and complex nature of bankruptcy filings. Predicting bankruptcy requires the model to distinguish not only firms in distress but also those likely to take legal action, which might occur after prolonged financial difficulties.

Interestingly, the AUC scores for bankruptcy prediction decline more slowly than those for financial distress over time. Specifically, the drop in AUC for bankruptcy predictions from $t-1$ to $t-3$ is relatively modest (2% and 5%, respectively), while for financial distress, this drop is more significant (12%). This result suggests that once a company is significantly distressed, bankruptcy is more predictable because firms tend to undergo a prolonged period of deterioration before filing for bankruptcy. In contrast, financial distress often occurs in the short term. As firms struggle over time, it becomes more challenging to predict distress accurately because they may oscillate between distressed and stable states. Our findings related to bankruptcy prediction confirm past studies' results. When considering classifier performance at $t-1$ models, our AUC results for precision and sensitivity are similar to Barboza et al. (2017), though our precision results are slightly higher.

Similarly, F1 scores are higher when analyzing financial distress rather than bankruptcies. This discrepancy might be because predictors other than financial, macroeconomic, and geographical variables can cause bankruptcy, and the variables now included in the model seem to predict financial distress better than bankruptcy.

Table 4**Performance measures for predicting financial distress at different time horizons ($t-1$, $t-2$, $t-3$).**

This table shows the performance metrics for predicting financial distress at different time horizons ($t-1$, $t-2$, $t-3$) using various ML models, including logit, k -NN, decision trees, Random Forest, AdaBoost, XGBoost, and neural network. The table shows the accuracy, precision, recall, F1 score, and AUC for each model at the three time points.

	$t-1$					$t-2$					$t-3$				
	Accuracy	Precision	Recall	F1	AUC	Accuracy	Precision	Recall	F1	AUC	Accuracy	Precision	Recall	F1	AUC
LOGIT (FULL)	0.947	0.045	0.929	0.086	0.978	0.877	0.015	0.842	0.030	0.937	0.803	0.008	0.842	0.015	0.893
LOGIT (STEPWISE)	0.947	0.045	0.929	0.085	0.977	0.877	0.015	0.842	0.029	0.937	0.802	0.008	0.844	0.015	0.893
K-NN	0.966	0.064	0.850	0.118	0.934	0.940	0.025	0.621	0.047	0.824	0.930	0.009	0.375	0.018	0.699
DECISION TREES	0.985	0.101	0.591	0.173	0.789	0.972	0.034	0.381	0.062	0.677	0.965	0.012	0.232	0.023	0.599
RANDOM FOREST	0.995	0.317	0.786	0.452	0.988	0.993	0.182	0.481	0.264	0.951	0.998	0.106	0.043	0.061	0.892
ADABOOST	0.992	0.219	0.794	0.343	0.986	0.99	0.117	0.492	0.19	0.941	0.989	0.03	0.165	0.051	0.879
XGBOOST	0.997	0.46	0.718	0.561	0.991	0.996	0.281	0.396	0.329	0.949	0.997	0.092	0.079	0.085	0.901
NEURAL NETWORK	0.932	0.028	0.729	0.055	0.928	0.95	0.03	0.615	0.057	0.904	0.925	0.018	0.757	0.036	0.933

Table 5**Performance measures for predicting bankruptcy at different time horizons ($t-1$, $t-2$, $t-3$).**

This table shows the performance metrics for predicting bankruptcy at different time horizons ($t-1$, $t-2$, $t-3$) using various ML models, including logit, k -NN, decision trees, Random Forest, AdaBoost, XGBoost, and neural network. The table shows the accuracy, precision, recall, F1 score, and AUC for each model at the three time points.

	$t-1$					$t-2$					$t-3$				
	Accuracy	Precision	Recall	F1	AUC	Accuracy	Precision	Recall	F1	AUC	Accuracy	Precision	Recall	F1	AUC
LOGIT (FULL)	0.924	0.004	0.771	0.008	0.912	0.883	0.006	0.799	0.012	0.916	0.828	0.005	0.801	0.010	0.881
LOGIT (STEPWISE)	0.923	0.004	0.771	0.008	0.913	0.883	0.006	0.795	0.012	0.914	0.828	0.005	0.795	0.010	0.882
K-NN	0.985	0.01	0.415	0.019	0.736	0.968	0.012	0.458	0.024	0.768	0.968	0.012	0.45	0.024	0.747
DECISION TREES	0.995	0.015	0.208	0.027	0.601	0.987	0.012	0.172	0.023	0.58	0.981	0.008	0.15	0.016	0.566
RANDOM FOREST	0.999	0.124	0.35	0.183	0.938	0.998	0.153	0.327	0.208	0.948	0.998	0.171	0.206	0.187	0.919
ADABOOST	0.994	0.037	0.488	0.068	0.93	0.992	0.046	0.419	0.083	0.911	0.992	0.039	0.25	0.068	0.879
XGBOOST	0.999	0.289	0.301	0.295	0.953	0.999	0.248	0.302	0.273	0.956	0.999	0.318	0.219	0.259	0.922
NEURAL NETWORK	0.943	0.004	0.561	0.008	0.832	0.926	0.007	0.649	0.015	0.878	0.896	0.007	0.638	0.013	0.878

6.2 Feature differences between bankruptcy and financial distress models (H2)

The objective of this section is to test whether the models can discriminate between financially distressed firms and bankrupt firms by testing the explanatory power of the most relevant features. If a company filed a financial distress procedure in any given year, then the observation in that year is categorized with 1 in the financial distress prediction model. Similarly, if a company filed a bankruptcy procedure in any given year, then the observation in that year is categorized with 1 in the bankruptcy prediction model. The process is reiterated for observations two years before bankruptcy ($t-2$) and three years before bankruptcy ($t-3$). Financially healthy firms are assigned the value of 0. Based on previous results (Section 6.1), we focus on the findings from implementing XGBoost and Random Forest models to identify the main determinants of financial distress and bankruptcy. We then compare these results with those obtained from the logit model, which is less of a “black box.”

6.2.1 Financial data

Table 6 shows the resulting parameters, *chi*-squared statistics, and *p*-values of the logit model for the year $t-1$.⁸ We entered financial data one year before the firm filed a bankruptcy or financial distress procedure. In addition, we developed these models using a backward selection approach, as in Jones et al. (2017). The criteria used to select the predictor to eliminate from the regression is the Bayesian Information Criterion (BIC). As in Shumway (2001), the *chi*-squared statistic is computed for each predictor and averaged across years.

Overall, most predictor estimates show a predicted sign. In terms of profitability, a high ROE (R5 & R6) decreases the likelihood of financial distress and bankruptcy, as well as higher net income to EBIT (R26) and EBITDA to current liabilities (R25).

In terms of efficiency, a high amount of fixed assets (R29) or cash (R28) held relative to sales increases the likelihood of both financial distress and bankruptcy. In contrast, a high ratio of payables to expenditure (e.g., R42 and R33—the two ratios differ slightly depending on what is considered an expense) signals that a firm is experiencing financial distress. In terms of solvency, highly leveraged firms are more subject to default risk. When considering liabilities to sales (R43) and total debt to total assets (R62 and R63), excessive debt increases the probability of default. Interestingly, only the financial distress prediction model has significant interest expense ratio estimates (R52 and R54). Low asset liquidity also impacts the probability of distress. According to the logit model results, higher cash and a quick ratio (R67, R69) are associated with a lower likelihood of distress in the following years. In addition, excessive receivables to current assets (R65) seem to increase the probability of both

⁸ The same logit model has also been estimated using financial data from two ($t-2$) and three ($t-3$) years before the company filed the financial distress or bankruptcy procedure (see Tables 1B and 2B in the Appendix B). Most of the regressors retain the same sign and statistical significance level, indicating marked consistency over time.

bankruptcy and financial distress and individuate eventual problems in the firm turning receivables into cash.

Table 6

Logit model results at time $t-1$.

This table shows the estimates for financial distress (FDP) and bankruptcy prediction (BKP) (year $t-1$). Significance levels: '10%, *5%, **1%, ***0.1%.

Code	Variable	FDP	Chi-sq	S	BKP	Chi-sq	S
(Intercept)		-15.074	2665.45	***	-1.999	11.91	
R2	Cash flow/Short-term debt	-1.009	66.03	**	1.350	39.47	*
R3	Interest expense/Cash flow	-0.573	283.42	***			
R4	Cash flow/Sales	-0.737	165.93	***			
R5	After-tax return on total equity (ROE)	-0.793	329.73	***	-0.349	9.69	
R6	After-tax return on common equity (ROE)	-0.370	103.72	***	-0.503	36.67	*
R7	EBITDA/Sales				-0.346	10.04	
R8	Profit margin after depreciation/Sales				1.949	20.02	'
R9	Profit margin before depreciation/Sales				-2.657	39.37	*
R10	Return on investment (ROI)	0.334	45.17	*	0.757	40.50	*
R11	Service expenses/Sales	-1.094	198.67	***			
R14	Labor expenses/EBITDA	-0.098	13.37		-0.264	15.34	
R15	Added value/Sales	0.292	41.77	*	0.867	69.03	**
R16	Effective tax rate	-0.703	227.73	***	-1.467	211.78	***
R17	Net profit margin ROS				0.551	14.12	
R18	Third use assets/Sales	-1.646	209.68	***	-1.232	29.50	*
R19	Return on capital (Invested)				0.375	11.09	
R21	Pre-tax profit margin	0.365	23.82	'			
R22	Return on assets	-2.922	331.72	***	2.541	91.93	***
R25	EBITDA/Current liabilities	0.623	20.57	'	-1.870	85.99	***
R26	Net income/EBIT	0.393	149.32	***	1.049	154.38	***
R27	Amortization and depreciation/Sales	0.532	32.97	*			
R28	Cash and cash equivalents/Sales				-0.930	60.88	**
R29	Fixed assets/Sales (Long-term assets turnover)	-0.291	13.36		-1.128	53.10	**
R30	Net working capital/Sales	-0.165	15.88				
R33	Payables turnover	0.686	192.09	***	0.686	49.91	**
R35	Stockholders' equity/Sales				1.634	91.04	***
R36	Total assets/Sales (with the value of production)	0.598	61.54	**	5.192	196.83	***
R37	Current assets/Sales (Current assets turnover)				-4.724	173.06	***
R40	Inventory over sales (inv. inventory turnover) (var)	0.549	176.36	***			
R41	Receivables over sales (inv. receivable turnover) (var)				-0.575	48.59	**
R42	Payables over expenditures (inv. payables turnover) (var)	0.766	274.00	***	1.023	131.02	***
R43	Total liabilities/Sales	2.379	929.77	***	1.733	122.68	***
R46	Common equity/Invested capital	1.068	297.76	***	1.123	71.81	**
R47	Total debt/Invested capital	-0.261	44.97	*	-0.315	13.36	
R49	Fixed assets/Total assets	1.199	66.83	**			
R51	Total assets/Total equity (Net worth)	-1.378	830.30	***	-0.715	61.93	**
R52	Interest expense/Long-term debt (cost of debt)	0.170	24.68	'	0.220	12.45	
R54	Interest expense/Sales	1.035	145.91	***			
R56	Total equity/Total assets				-3.648	149.42	***
R58	Debt/Equity	0.340	61.21	**			
R60	Total debt/Common equity	1.045	291.81	***	0.519	16.80	
R61	Total debt/EBITDA	-0.249	19.20	'	-0.994	74.51	**
R62	Total debt/Total assets	1.968	522.29	***	2.105	137.53	***
R63	Total Liabilities/Total assets	9.522	1540.25	***	-2.949	36.38	*
R65	Receivables/Current assets	1.114	146.02	***	2.515	86.24	***
R67	Cash ratio				-2.879	71.57	**
R68	Current ratio	0.985	52.27	**	-1.731	19.06	'
R69	Quick ratio (acid test)				-1.028	12.77	
R70	Current indebtedness ratio	1.288	93.16	***	-4.062	202.38	***
R72	Cash balance/Total liabilities	2.405	124.46	***	5.416	158.12	***
R73	Cash and cash equivalents/Total assets	-3.035	182.94	***	1.703	18.95	'
R78	Current assets/Total assets	-0.889	16.06		7.487	310.44	***
R79	Sales per employee				0.026	55.93	**
CPI	Consumer prices: all items	-0.457	112.47	***			
LTI	Long-term interest rate	0.789	431.36	***	-0.439	39.60	*
IPI	Industrial production, s.a.				0.331	107.62	***
RTV	Retail trade (volume), s.a.	-0.144	69.12	**			

TER	Total employment: all persons, s.a.	0.476	127.38	***	-0.929	61.17	**
AGR	Agriculture & Mining, Quarrying, and Oil and Gas Extraction				0.959	28.10	*
UTI	Utilities	0.683	64.43	**	1.388	72.04	**
CON	Construction	0.556	120.47	***			
MAN	Manufacturing	0.419	86.24	***			
TRA	Wholesale + retail trade	0.298	31.18	*	-0.455	23.83	'
TRS	Transportation & warehousing	0.367	20.84	'	-1.391	51.65	**
NE	Northeast macro-region	0.150	13.63		0.628	60.78	**
C	Center macro-region	0.393	87.65	***			
SI	South-Islands macro-region	0.448	107.20	***	0.524	33.37	*

Previous studies established that despite algorithms such as XGBoost and Random Forest performing substantially better than simple classifiers, their interpretation in practical settings is more complex. We used the SHAP method to explain how the prediction given by our best model, the XGBoost, is distributed among each feature in the dataset. The SHAP importance score of each attribute indicates how valuable each predictor is in constructing the decision trees. Larger Shapely values (SVs) mean that these features are more important than others.

For each model at a different time horizon previously analyzed, we considered the 20 most valuable predictors and combined them in a summary plot to compare their performance predictivity.⁹ The density scatter plot is composed of n points as the number of observations in the dataset, while the color of each point identifies the value of the ratio: purple high; blue low. The x -axis illustrates the magnitude of the computed SVs—the higher the absolute value of the SVs, the higher the magnitude of the variable in explaining the outcome, and the SVs are ranked in Figure 3. For example, concerning the model BKP $t-1$ in Figure 3, feature R16 is ranked higher than R62, meaning it generally exhibits a higher impact in classifying a firm as bankrupt. However, for some observations where R16 is extremely low (blue), feature R62 has a higher impact on the classification than R16. R16 affects the classification of many observations by a small magnitude, while R62 affects the classification of a small number of observations by a large magnitude.

Comparing the three financial distress prediction models made it possible to identify eight common predictors. The most relevant and constant predictors throughout time are as follows: net profit margin (R17), payables to expenditures (R42), total liabilities to sales (R43), interest expense to long-term debt (R52), and interest expenses to sales (R54). It is worth noticing the high SV and financial distress probability when profit margins (R17) are low or payables to expenditures (R42) are high. In terms of solvency, high values of interest expense to long-term debt (R52) or sales (R54) and high liabilities to sales considerably increase the probability of default.

⁹ Estimates for the remaining variables are available upon request.

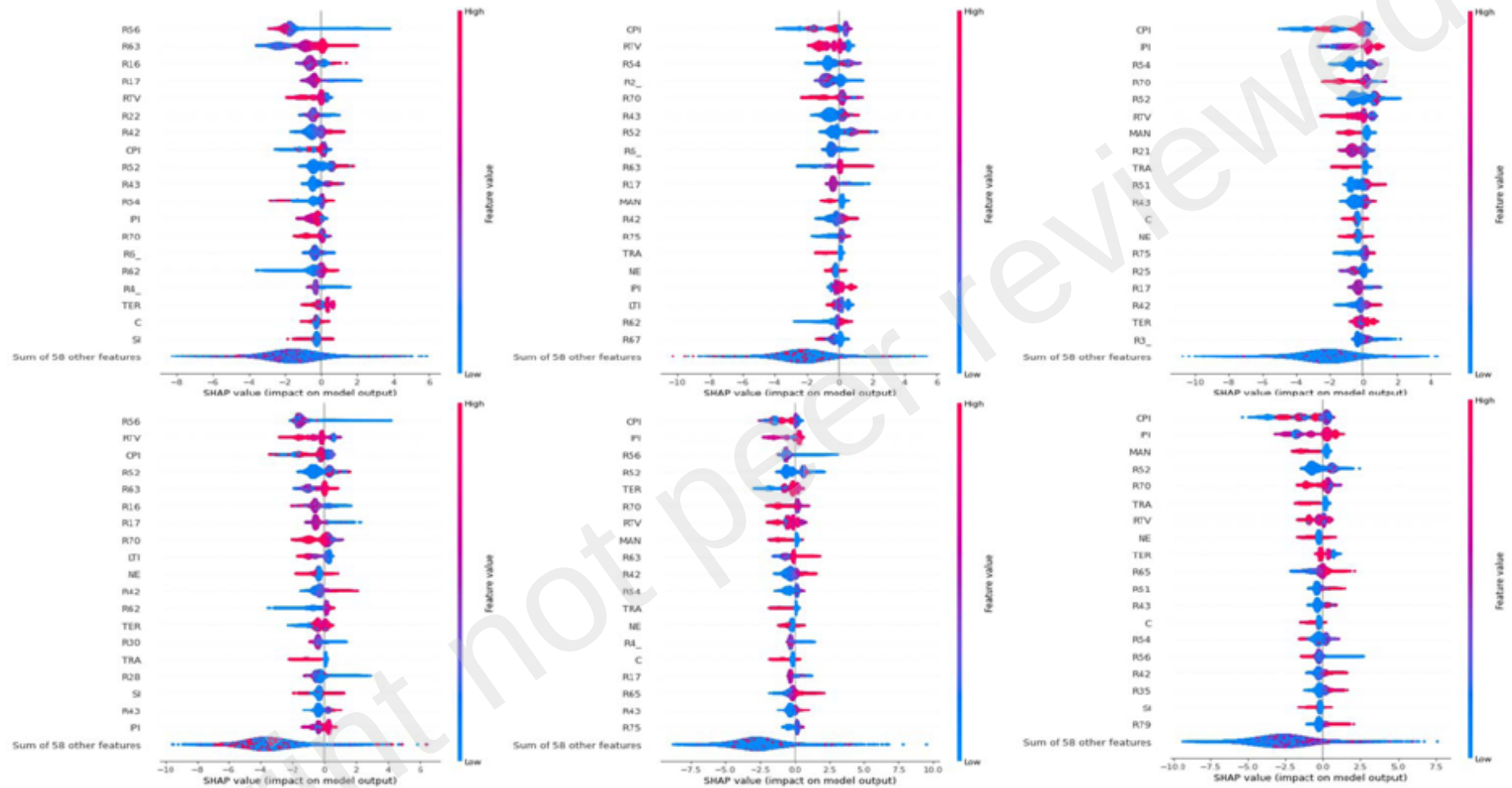


Fig. 3. Importance of features according to XGBoost model.

This figure shows the SHAP values obtained using the XGBoost. From left to right, the first row shows the financial distress probability (FDP) at time $t-1$, $t-2$, and $t-3$. The second row shows the bankruptcy probability (BKP) at time $t-1$, BKP $t-2$, BKP $t-3$.

In contrast, a different subset of predictors appears among the 20 more relevant for the case of bankruptcy. This subset includes payables to expenditures (R42), total liabilities to sales (R43), interest expense to long-term debt (R52), total equity to total assets (R56), and the current indebtedness ratio (R70). Their interpretation is analog to the case of financial distress. However, different from financial distress prediction, it seems that leverage, measured as equity to total assets, is one of the relevant predictors. Significantly, low equity to total assets leads to high SVs, which means a higher probability of bankruptcy. Moreover, a low current debt (indebtedness) ratio seems to lessen the probability of bankruptcy.

Overall, according to this analysis, solvency variables are essential predictors of both financial distress and bankruptcy. However, profitability ratios, such as profit margin, are consistently found to be valuable predictors of financial distress.

6.2.2 Macroeconomic data

Regarding macroeconomic predictors, the CPI, retail trade growth (RTV), and industrial production index (IPI) consistently predict both financial distress and bankruptcy. This finding reinforces the need to include macroeconomic predictors in such models.

The inclusion of macroeconomic predictors considerably improves performance. Variable importance methods for both XGBoost and Random Forest demonstrated that CPI, RTV, and IPI positively contributed to the model's performance. The effects of each predictor on the likelihood of bankruptcy seem to cohere with other studies (see Section 3). The CPI indicates a general increase in prices and inflation. According to the estimated models, inflation growth increases the likelihood of financial distress and bankruptcy. Generally, inflation growth results from sustained consumption and stimulates investment in firms, reducing financial distress probability. However, the models are trained in a period characterized by low inflation (2013–2020). Other studies ((Mare 2015) have found that overly high inflation may be related to poor macroeconomic conditions and consequently increase the likelihood of bankruptcy. RTV, IPI, and TER are also variables that tend to mirror the conditions of the economy. An increase in the TER seems to signify a healthy economy and fewer bankruptcies. RTV seems to have inconsistent signs among the models, while the IPI seems to decrease the likelihood of financial distress. Lastly, the growth of the long-term interest rate (LTI) seems to decrease the probability of bankruptcies. As mentioned by other authors, growth in the interest rate spread and a positively sloped yield curve are related to positive expectations for the future, while a downward-sloping curve usually indicates a recession.

6.2.3 Industry sector data

We researched the type of industrial activity of firms and their geographical location to observe the possible effects of these features on the crises firms face. Prior literature has included these variables in

their models, but the results are ambiguous (Becchetti and Sierra 2003; Amendola et al. 2011). This ambiguity may be due to how a specific variable is encoded into a model. For example, the industry classification of Italian firms usually depends on a six-digit code called ATECO, which represents the primary activity the company is pursuing. However, the company may change the nature of its activities throughout its life, or its secondary activities may belong to a different sector and be relevant.

All the firms included in the sample were classified according to their ATECO code (see Section 5). As shown in Table 6 and relative Tables 1B and 2B in the Appendix B, only the manufacturing sector demonstrates a significantly higher probability of becoming financially distressed at a 1% confidence level, while the utilities and agriculture sectors have a significantly higher likelihood of going bankrupt. However, according to both SHAP values and feature importance methodologies, not one of the industry sector dummy variables was steadily in the 20 most important predictors (see Figure 3). Nonetheless, when analyzing variable importance, it is noticeable that in four models out of six, the manufacturing category exhibits a higher probability of bankruptcy than the other groups. Overall, industry classification may give misleading results. Other authors have noticed that most firms slightly change their activities during their lives, or the ATECO classification may be incorrectly classified (Perboli and Arabnezhad 2021).

6.2.4 Macro-regions

Controls for the location of the legal headquarters in the four macro-regions (i.e., Northwest, Northeast, Center, and South-Islands) show that they are not statistically significant in all the financial distress prediction models from $t-1$ to $t-3$. Also, when considering SVs, these macro-regions are found to be marginally important for the model. When comparing the bankruptcy prediction models, only the Northeast area seems to have predictive power, though the signs of the coefficients are not stable across the models. It is possible to conclude that the geographical location intended for the legal headquarters of the firm has an insignificant impact on financial distress and bankruptcy probabilities. This may be for many reasons. For example, a firm may register a company in a region and conduct business elsewhere. In addition, the sample analyzed here includes large companies that conduct their activities in the whole country or even abroad, reducing the relevance of this classification. Overall, our results confirm the findings of previous studies (e.g., Perboli and Arabnezhad (2021)).

6.3 The impact of COVID-19 on predictive performance (H3)

The previous analysis, spanning from 2013 to 2020, includes data from 2020, the year the World Health Organization declared COVID-19 a global pandemic. Given the widespread effects of the pandemic and the accompanying governmental regulations, it is plausible that these factors could have significantly influenced bankruptcy and financial distress filings in 2020, potentially diminishing the accuracy of our classifiers. Specifically, the early months of 2020 saw a notable reduction in bankruptcy

and insolvency filings due to the suspension of business operations and court activities. At the same time, many companies experienced rapid liquidity declines and found it hard to repay their debts. Therefore, the inclusion of 2020 data could either enhance or detract from the predictive capabilities of the classifiers.

To test hypothesis H3, we performed a non-parametric comparison of the areas under the ROC curves. Using the method proposed by DeLong et al. (1988), we estimated the mean AUC and the corresponding confidence intervals. We then conducted two rounds of testing: the first round used the full dataset; the second excluded the 2020 data. Both rounds employed the same procedure: we split the dataset into 70% training and 30% test sets, applied SMOTE to address the class imbalance, and then trained the classifiers. Finally, we compared the ROC curves and their confidence intervals to detect any statistical differences between the two rounds.

The results (presented in Tables 7 and 8) suggest that including 2020 data did not significantly affect classifier performance. In nearly all cases, the 95% confidence intervals of the AUC values overlap, indicating that including 2020 data did not notably alter the classifiers' effectiveness. The only exception is a slight, statistically insignificant decrease in the AUC score when the 2020 data were excluded. Some classifiers, such as k -NN, show marginally lower performance due to the reduced training data. These findings are in line with the results in Tables 4 and 5, providing validation for the robustness of our models. This validation process is crucial, as resampling the dataset can sometimes influence the results. However, as demonstrated by the confidence intervals and mean AUC in Tables 7 and 8, including 2020 data did not lead to significant changes. XGBoost remains the top-performing classifier, achieving the highest AUC, followed by Random Forest. AdaBoost, logistic regression, and Neural Network are the third most effective models, while k -NN and decision trees show the least effectiveness.

Table 7
Mean AUC and confidence intervals for financial distress prediction at different time horizons (at $t-1$, $t-2$, $t-3$).

Model	$t-1$						$t-2$						$t-3$					
	2013–2020			2013–2019			2013–2020			2013–2019			2013–2020			2013–2019		
	Mean AUC	LB	UB	Mean AUC	LB	UB	Mean AUC	LB	UB	Mean AUC	LB	UB	Mean AUC	LB	UB	Mean AUC	LB	UB
Logit	0.978	0.972	0.983	0.979	0.974	0.984	0.937	0.929	0.945	0.938	0.929	0.947	0.894	0.882	0.905	0.870	0.855	0.885
k -NN	0.929	0.918	0.940	0.925	0.913	0.936	0.828	0.810	0.845	0.809	0.791	0.826	0.717	0.695	0.739	0.715	0.694	0.735
Decision trees	0.815	0.798	0.831	0.818	0.802	0.834	0.694	0.676	0.712	0.710	0.692	0.728	0.627	0.607	0.647	0.640	0.621	0.659
Random Forest	0.956	0.950	0.963	0.911	0.899	0.922	0.936	0.923	0.948	0.936	0.927	0.944	0.960	0.953	0.967	0.951	0.941	0.960
AdaBoost	0.982	0.977	0.987	0.979	0.973	0.985	0.939	0.930	0.948	0.939	0.921	0.950	0.883	0.871	0.895	0.870	0.857	0.883
XGBoost	0.988	0.984	0.992	0.986	0.981	0.990	0.950	0.942	0.958	0.942	0.934	0.950	0.890	0.876	0.903	0.883	0.871	0.896
Neural network	0.991	0.988	0.993	0.988	0.984	0.991	0.950	0.942	0.958	0.944	0.936	0.951	0.900	0.888	0.913	0.887	0.876	0.899

Table 8
Mean AUC and confidence intervals for bankruptcy prediction at different time horizons (at $t-1$, $t-2$, $t-3$).

Model	$t-1$						$t-2$						$t-3$					
	2013–2020			2013–2019			2013–2020			2013–2019			2013–2020			2013–2019		
	Mean AUC	LB	UB	Mean AUC	LB	UB	Mean AUC	LB	UB	Mean AUC	LB	UB	Mean AUC	LB	UB	Mean AUC	LB	UB
Logit	0.914	0.880	0.948	0.891	0.849	0.932	0.916	0.897	0.935	0.919	0.899	0.938	0.881	0.863	0.900	0.858	0.835	0.880
k -NN	0.717	0.669	0.765	0.737	0.688	0.786	0.773	0.743	0.803	0.732	0.703	0.762	0.694	0.666	0.722	0.699	0.672	0.727
Decision trees	0.600	0.561	0.639	0.608	0.567	0.648	0.617	0.590	0.643	0.602	0.577	0.626	0.590	0.567	0.613	0.594	0.571	0.616
Random Forest	0.910	0.882	0.939	0.883	0.850	0.916	0.919	0.902	0.936	0.854	0.829	0.880	0.926	0.911	0.942	0.921	0.909	0.934
AdaBoost	0.921	0.890	0.952	0.910	0.877	0.944	0.918	0.901	0.936	0.907	0.888	0.926	0.887	0.868	0.905	0.883	0.865	0.901
XGBoost	0.941	0.913	0.969	0.920	0.886	0.953	0.946	0.929	0.963	0.938	0.921	0.955	0.916	0.899	0.933	0.907	0.889	0.925
Neural network	0.951	0.923	0.978	0.931	0.902	0.960	0.954	0.9420	0.966	0.945	0.932	0.959	0.930	0.917	0.942	0.924	0.917	0.937

7. Robustness check: Random Forest feature importance with permutation feature importance

This section validates the results obtained in Section 6 using a slightly different method, permutation feature importance (PFI), to assess the critical variables in the models. We applied PFI to Random Forest (as first applied by Breiman, 2001) because RF was one of the best performers individuated in the previous section, and PFI allows us to understand the interactions of features that contribute to the predictive accuracy of the model. PFI evaluates the importance of a feature by measuring the increase in prediction error when the feature is randomly permuted. If the prediction error remains unchanged after permuting a feature, the feature is irrelevant to the model (Breiman 2001).

PFI confirms many of the predictors identified earlier while also uncovering important new variables. Among the 20 most relevant features for predicting financial distress across all three periods, several accounting ratios emerge as critical, including interest expense to cash flow (R3), after-tax return to total equity (R5), after-tax return to common equity (R6), labor expenses to EBITDA (R14), return on invested capital (R19), pre-tax return to net operating assets (R20), fixed assets to total equity (R48), total assets to total equity (R51), interest expense to EBITDA (R53), LTI, growth in retail trade volume (RTV), and TER. These accounting ratios are primarily associated with profitability and solvency, highlighting their relevance for financial distress prediction.

An interesting finding is the variation in the importance of these indicators over the three years leading to the event, suggesting that the predictive power of certain features evolves over time. In period $t-1$, solvency-related variables emerge as the primary predictors of financial distress (R48, R51, R56), whereas in period $t-2$, profitability variables (R3, R5, R17) take precedence. By $t-3$, however, macroeconomic indicators become the most significant (RTV, IPI, CPI), suggesting that financial ratios lose some predictive power for financial distress as the time horizon extends beyond the short term.

According to PFI, the financial variables that predict financial distress are also highly relevant for predicting bankruptcy. In addition to the previously identified predictors, we found new significant variables, including the debt-to-equity ratio (R58), current indebtedness ratio (R70), and cash and cash equivalents to total assets ratio (R73). Unlike financial distress, liquidity-based predictors also play a role in bankruptcy prediction; however, solvency and profitability ratios remain the most important.

Based on this validation, we conclude that the key determinants for predicting both bankruptcy and financial distress are much the same, with solvency ratios being the most influential, followed by profitability measures. As noted by Jones et al. (2017), the results of ML algorithms are particularly reliable due to their ability to handle high dimensionality and correlations among variables. Furthermore,

the rankings presented in Figures 3 and 4 are less prone to analyst bias, making the model's findings more objective and consistent.¹⁰

¹⁰ Figures on the relevance of all variables are available upon request. Table 3B in Appendix B shows the PFI scores and confidence intervals according to Random Forest.

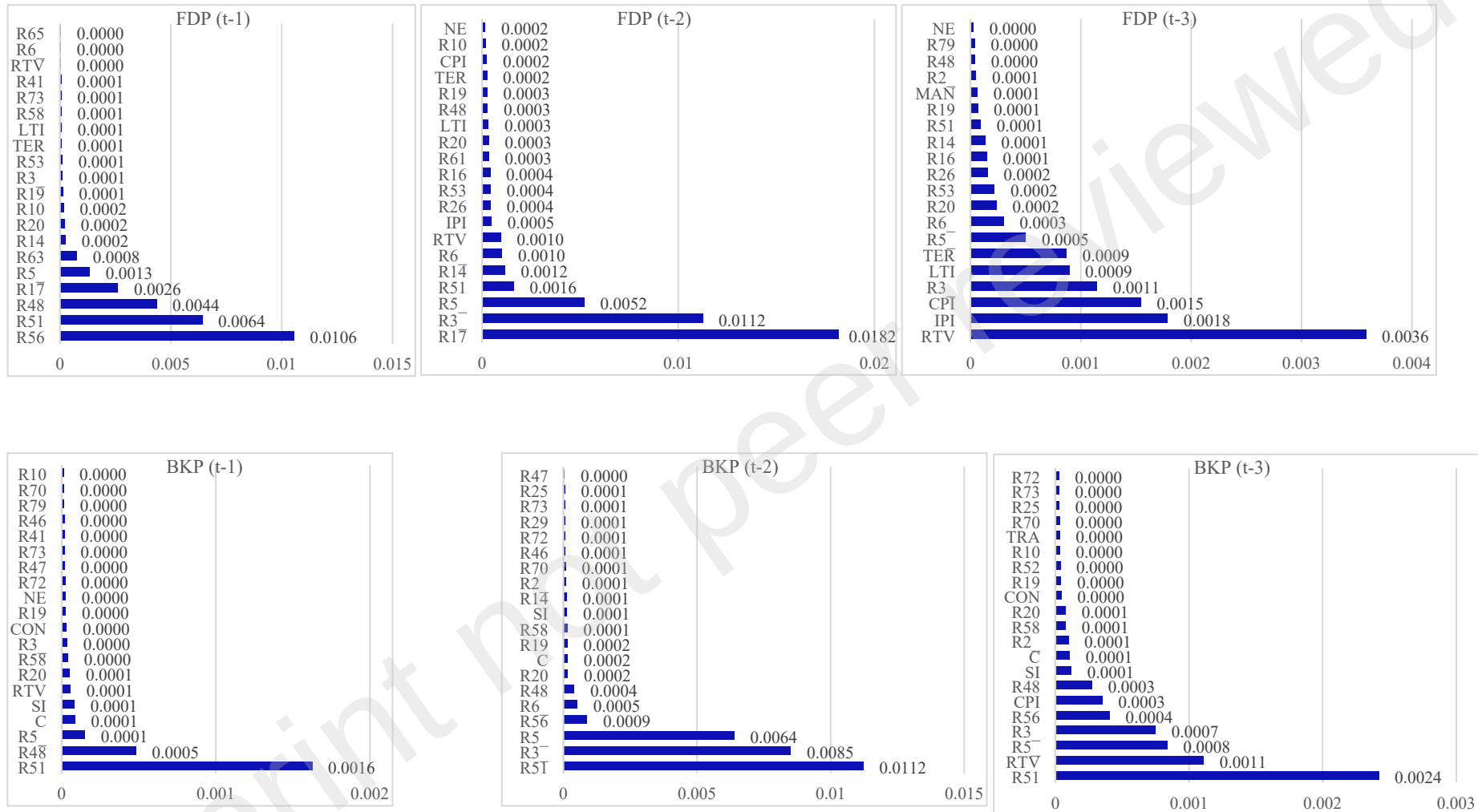


Fig. 4. Importance of features according to Random Forest model.

This figure shows the predictors for the financial distress probability (FDP, first row) and bankruptcy probability predictors (BKP, second row) at $t-1$, $t-2$ and $t-3$.

8. Conclusions

This paper sets out to identify the main predictors of firm financial distress and bankruptcy, focusing on determining their relative importance. The novelty of our approach lies in examining the differences in determinants for predicting financially distressed firms versus bankrupt firms. We applied this framework to a sample of Italian SMEs from 2013 to 2020, a setting well-suited to address our research question given the specific legal context. Financially distressed firms are classified as those filing debt restructuring agreements or undergoing other minor procedures, while bankrupt firms are categorized as failures.

The results highlight several key contributions to the existing literature on corporate failure prediction. First, this study demonstrates that including macroeconomic variables significantly enhances the predictive accuracy of financial distress models. While previous research has primarily relied on firm-level financial ratios to forecast bankruptcy, we show that macroeconomic factors such as GDP growth, interest rates, and inflation provide valuable additional information, particularly when combined with machine learning algorithms. This finding aligns with prior studies emphasizing the role of economic conditions in firm survival but extends the literature by confirming the robustness of macroeconomic predictors across multiple classification methods.

Second, this paper advances the understanding of financial distress by distinguishing it from bankruptcy based on legal classifications rather than arbitrary financial thresholds. Many prior studies have treated these concepts interchangeably, yet our findings indicate that they follow different predictive patterns. Financial distress, which may be reversible through restructuring or credit interventions, is inherently more challenging to predict in the medium term due to the fluctuating financial conditions of distressed firms. In contrast, bankruptcy follows a more deterministic trajectory, as firms experiencing severe financial deterioration proceed toward insolvency typically through judicial procedures. This differentiation enhances the applicability of financial distress models to real-world policymaking and regulatory frameworks, particularly in jurisdictions where insolvency proceedings differ significantly.

Third, our empirical analysis demonstrates the effectiveness of machine learning techniques in improving bankruptcy and financial distress prediction. Ensemble methods, particularly Random Forest, XGBoost, and AdaBoost, outperform traditional econometric models in most cases, achieving higher predictive accuracy across multiple time horizons. However, our results also reveal that logit models remain highly competitive, suggesting that simpler models can still be useful, particularly for interpretability and regulatory applications. These findings contribute to the ongoing debate regarding the superiority of machine learning over traditional statistical methods, highlighting the importance of balancing predictive power with model transparency and ease of implementation.

A further important insight emerges from the robustness of our models during economic shocks, such as the COVID-19 pandemic. Despite the severe financial disruption caused by the pandemic, our predictive framework remains valid, capturing financial distress trends even in an environment of widespread uncertainty. The unexpected decline in bankruptcy rates during 2020 and 2021, largely driven by government support measures such as loan moratoria and state-backed credit guarantees, suggests that while temporary interventions can delay insolvency, they do not necessarily resolve underlying financial weaknesses. This underscores the need for policymakers to consider long-term financial health beyond short-term crisis response, as businesses may still face heightened bankruptcy risk once emergency support measures are withdrawn.

The findings of this study have broader implications beyond the Italian context. Other countries can draw valuable lessons from our results in designing financial distress prediction models and regulatory frameworks. One key takeaway is the importance of incorporating macroeconomic indicators into early warning systems, as firm-level financial data alone may not fully capture the external conditions contributing to financial distress. Additionally, regulatory bodies and financial institutions should tailor prediction models to align with national bankruptcy laws, as insolvency proceedings vary significantly across jurisdictions. Our results also reinforce the potential of machine learning to enhance credit risk assessment, particularly for SMEs, which often face more obstacles in accessing financing and are more vulnerable to economic downturns.

Despite the contributions of this study, several limitations suggest promising directions for future research. One key opportunity is to apply the predictive framework to different countries, enabling cross-country comparisons that could help refine models tailored to diverse legal, regulatory, and economic environments. Moreover, although our models demonstrate strong predictive performance, incorporating additional data sources—such as payment behaviors, corporate governance structures, or sentiment analysis from financial news—could further enhance their accuracy and robustness.

Additionally, a more detailed sectoral analysis could uncover whether financial distress manifests differently across industries, offering valuable insights into sector-specific risk factors. Furthermore, integrating emerging machine learning classifiers and advanced feature importance techniques could elevate model performance. For instance, incorporating methods like convolutional neural networks or graph neural networks—capable of processing both textual and numerical data innovatively—may provide deeper insights into the relationships between predictor variables, thereby improving predictive accuracy.

Although the inclusion of data from 2020 did not significantly affect classifier performance, it is worth noting that our dataset ends that year. Future studies incorporating more recent data are essential to improve our understanding of the long-term effects of the COVID-19 pandemic on financial behavior and market conditions. As the economic impact of the pandemic continues to evolve, the predictability

of financial distress and default may shift, and additional years of data may reveal emerging patterns not yet captured in the current analysis.

To conclude, this study contributes to the growing field of financial distress and bankruptcy prediction by demonstrating the effectiveness of machine learning techniques, emphasizing the importance of macroeconomic indicators, and aligning predictive models with regulatory frameworks. The results offer investors, policymakers, and financial institutions valuable insights into the potential of predictive analytics to improve financial stability and inform early intervention strategies. While machine learning enhances predictive capabilities, regulatory alignment and economic context remain critical factors in developing reliable and actionable early warning systems.

Appendix A

Table 1A.
Main firm's financial distress and bankruptcy determinants.

Code	Description	Source in literature
(1) Profitability items/ratios		
R2	Cash flow/Short-term debt	Min and Lee (2005)
R3	Interest expense /Cash flow	Min & Lee (2005)
R4	Cash flow/Sales	Amendola et al. (2011)
R5	After-tax return on total equity (ROE)	Amendola et al. (2011); Becchetti & Sierra (2003)
R6	After-tax return on common equity (ROE)	Amendola et al. (2011)
R7	EBITDA/Sales	Amendola et al. (2011); Ciampi & Gordini (2013)
R8	Profit margin after depreciation/Sales	Amendola et al. (2011)
R9	Profit margin before depreciation/Sales	Zięba (2016)
R10	Return on investment (ROI)	Ciampi & Gordini (2013)
R11	Service expenses/Sales	Own
R12	Raw material expenses/Sales	Own
R14	Labor expenses/EBITDA	Own
R15	Added value/Sales	Ciampi (2015)
R16	Effective tax rate	Lin and McClean (2001)
R17	Net profit margin	Amendola et al (2011)
R18	Third use assets/Sales	Own
R19	Return on capital (Invested)	Lin & McLean (2001)
R20	Pre-tax return on net operating assets	Lin & McLean (2001)
R21	Pre-tax profit margin	Zięba (2016)
R22	Return on assets	Amendola et al. (2011); Zięba (2016)
R25	EBITDA/Current liabilities	Own
R26	Net income/EBIT	Amendola et al. (2011)

(2) Efficiency items/ratios

R27	Amortization and depreciation/Sales	Altman et al. (2010)
R28	Cash and cash equivalents/Sales	Own
R29	Fixed assets/Sales (long-term assets turnover)	Amendola et al. (2011)
R30	Net working capital/Sales	Alfaro et al (2008)
R32	Inventory over sales (with value of production)	Amendola et al. (2011)
R33	Payables turnover	Amendola et al. (2011)
R35	Stockholders equity/Sales	Own
R36	Total assets/Sales (with value of production)	Alfaro et al (2008); Becchetti & Sierra (2003)
R37	Current assets/Sales (current assets turnover)	Amendola et al. (2011)
R38	Quick/Liquid assets over sales	Deakin (1972)
R40	Inventory over sales (inv. inventory turnover) (var)	Amendola et al. (2011)
R41	Receivables over sales (inv. receivable turnover) (var)	Amendola et al. (2011)
R42	Payables over expenditures (inv. payables turnover) (var)	Amendola et al (2011)

(3) Solvency items/ratios

R43	Total liabilities/Sales	Own
R46	Common equity/Invested capital	Own
R47	Total debt/Invested capital	
R48	Fixed assets/Total equity	Amendola et al. (2011); Zięba (2016)
R49	Fixed assets/Total assets	Own
R51	Total assets/Total equity (net worth)	Becchetti & Sierra (2003)
R52	Interest expense/Long-term debt (cost of debt)	Amendola et al. (2011)
R53	Interest expense/EBITDA	Ciampi (2015)
R54	Interest expense/Sales	Amendola et al (2011); Ciampi (2015), Becchetti & Sierra (2003)
R56	Total equity/Total assets	Amendola et al. (2011)
R58	Debt/equity	Ciampi (2015)
R60	Total debt/Common equity	Jones et al. (2017)
R61	Total debt/EBITDA	Ciampi (2015); Ciampi & Gordini (2013)

R62	Total debt/Total assets	Amendola et al. (2011); Jones et al. (2017), Becchetti & Sierra (2003)
R63	Total liabilities/Total assets	Zięba (2016)

(4) Liquidity items/ratios

R65	Receivables/Current assets	Own
R67	Cash ratio	Deakin (1972)
R68	Current ratio	Amendola et al (2011); Jones et al. (2017)
R69	Quick Ratio (Acid test)	Amendola et al. (2011)
R70	Current indebtedness ratio	Own
R72	Cash balance/Total liabilities	Deakin (1972)
R73	Cash and cash equivalents/Total assets	Deakin (1972)
R75	Inventories/Total assets	Own
R78	Current assets/Total assets	Amendola et al. (2011); Alfaro et al. (2008)

(5) Output items/ratios

R79	Sales per employee (in million €)	Ciampi (2015)
R80	Added value per employee (in million €)	Ciampi (2015)

(6) Macroeconomic ratios

CPI	Consumer prices: all items (% growth)	Bruneau et al. (2012); Mare (2012)
LTI	Long-term interest rate	Koopman & Lucas (2005)
IPI	Industrial production (% growth)	Bruneau et al. (2012)
RTV	Retail trade (% growth)	Bruneau et al. (2012)
TER	Total employment (% growth)	Carling et al. (2007) (similar to household expectations)

(7) Industry sector

AGR	Agriculture & Mining, Quarrying, and Oil and Gas Extraction
UTI	Utilities
CON	Construction
MAN	Manufacturing
TRA	Wholesale + Retail Trade

Table 2A.
Descriptive statistics of variables to predict financial distress and bankruptcy at time $t-1$, $t-2$, and $t-3$.

Panel A																		
Variable	Financially distressed firms									Bankrupt firms								
	$t-1$			$t-2$			$t-3$			$t-1$			$t-2$			$t-3$		
	N. Obs	mean	std	N. Obs	mean	std	N. Obs	mean	std	N. Obs	mean	std	N. Obs	mean	std	N. Obs	mean	std
R2	2625	-25.61%	28.63%	2268	-18%	27%	1761	-6%	22%	367	-7%	49%	857	-3%	38%	1,042	1%	34%
R3	2625	-13.33%	32.44%	2268	-3%	51%	1761	12%	62%	367	-5%	32%	857	1%	44%	1,042	7%	48%
R4	2625	-43.28%	42.77%	2268	-26%	41%	1761	-12%	35%	367	-26%	72%	857	-23%	62%	1,042	-16%	52%
R5	2625	40.78%	55.41%	2268	16%	65%	1761	-3%	50%	367	26%	57%	857	17%	55%	1,042	9%	49%
R6	2625	-80.68%	42.89%	2268	-62%	55%	1761	-32%	60%	367	-41%	75%	857	-38%	71%	1,042	-18%	74%
R7	2625	-33.82%	44.65%	2268	-18%	41%	1761	-5%	34%	367	-29%	69%	857	-26%	60%	1,042	-14%	52%
R8	2625	-45.35%	37.21%	2268	-27%	38%	1761	-11%	31%	367	-29%	60%	857	-26%	53%	1,042	-17%	45%
R9	2625	-34.41%	38.21%	2268	-18%	37%	1761	-4%	30%	367	-26%	59%	857	-22%	52%	1,042	-13%	45%
R10	2625	12.50%	49.50%	2268	7%	43%	1761	5%	29%	367	9%	55%	857	8%	47%	1,042	7%	41%
R11	2625	30.04%	37.29%	2268	27%	33%	1761	26%	29%	367	-8%	71%	857	1%	64%	1,042	11%	53%
R12	2625	28.96%	36.17%	2268	33%	35%	1761	34%	32%	367	-11%	67%	857	1%	62%	1,042	15%	54%
R14	2625	-29.24%	55.36%	2268	-5%	75%	1761	32%	76%	367	-13%	52%	857	-2%	65%	1,042	16%	66%
R15	2625	-8.71%	49.85%	2268	5%	44%	1761	16%	36%	367	11%	73%	857	12%	66%	1,042	12%	58%
R16	2625	0.75%	17.33%	2268	8%	31%	1761	19%	39%	367	1%	22%	857	8%	30%	1,042	15%	34%
R17	2625	-48.77%	44.96%	2268	-31%	41%	1761	-16%	34%	367	-29%	65%	857	-22%	58%	1,042	-17%	48%
R18	2625	3.06%	27.20%	2268	3%	24%	1761	2%	20%	367	-29%	55%	857	-20%	50%	1,042	-9%	41%
R19	2625	19.23%	63.15%	2268	4%	58%	1761	3%	38%	367	16%	57%	857	12%	51%	1,042	11%	46%
R20	2625	4.04%	66.21%	2268	-6%	54%	1761	-4%	33%	367	10%	59%	857	8%	50%	1,042	6%	42%
R21	2625	-48.49%	45.12%	2268	-30%	42%	1761	-15%	34%	367	-29%	65%	857	-21%	58%	1,042	-16%	48%
R22	2625	-32.08%	30.66%	2268	-18%	27%	1761	-6%	18%	367	-14%	47%	857	-12%	37%	1,042	-6%	29%
R25	2625	-19.64%	26.57%	2268	-12%	25%	1761	-1%	20%	367	-5%	47%	857	-2%	36%	1,042	4%	35%
R26	2625	73.74%	37.19%	2268	58%	50%	1761	38%	55%	367	66%	43%	857	57%	47%	1,042	45%	51%
R27	2625	16.77%	37.92%	2268	11%	31%	1761	7%	26%	367	-22%	61%	857	-14%	54%	1,042	-7%	43%
R28	2625	11.55%	35.77%	2268	5%	27%	1761	5%	22%	367	-15%	69%	857	-7%	59%	1,042	3%	51%
R29	2625	2.96%	29.11%	2268	4%	26%	1761	6%	24%	367	-26%	57%	857	-17%	52%	1,042	-7%	42%

R30	2625	-42.44%	59.60%	2268	-17%	57%	1761	0%	53%	367	-37%	74%	857	-25%	69%	1,042	-10%	63%
R32	2625	25.27%	40.14%	2268	29%	38%	1761	32%	36%	367	18%	39%	857	15%	38%	1,042	19%	35%
R33	2625	66.67%	39.44%	2268	60%	36%	1761	57%	37%	367	43%	54%	857	46%	53%	1,042	45%	50%
R35	2625	22.63%	33.97%	2268	17%	29%	1761	15%	27%	367	29%	39%	857	24%	39%	1,042	20%	35%
R36	2625	76.61%	38.51%	2268	77%	35%	1761	81%	28%	367	66%	45%	857	67%	46%	1,042	71%	36%
R37	2625	62.83%	39.46%	2268	63%	36%	1761	67%	30%	367	58%	45%	857	60%	46%	1,042	64%	37%
R38	2625	9.36%	63.30%	2268	28%	56%	1761	35%	51%	367	2%	79%	857	13%	74%	1,042	18%	69%
R40	2625	38.52%	45.06%	2268	42%	46%	1761	48%	44%	367	15%	75%	857	28%	65%	1,042	25%	60%
R41	2625	33.97%	38.70%	2268	32%	37%	1761	30%	36%	367	22%	64%	857	33%	55%	1,042	28%	53%
R42	2625	56.01%	45.05%	2268	44%	41%	1761	43%	35%	367	45%	67%	857	49%	57%	1,042	42%	55%
R43	2625	81.57%	42.02%	2268	78%	38%	1761	77%	34%	367	22%	91%	857	34%	82%	1,042	46%	67%
R46	2625	-3.18%	40.60%	2268	8%	38%	1761	14%	32%	367	5%	47%	857	7%	38%	1,042	8%	34%
R47	2625	-9.52%	79.61%	2268	30%	76%	1761	53%	56%	367	-1%	75%	857	14%	72%	1,042	24%	67%
R48	2625	-19.44%	57.10%	2268	16%	68%	1761	45%	56%	367	-4%	45%	857	11%	49%	1,042	24%	50%
R49	2625	25.14%	23.62%	2268	22%	22%	1761	21%	21%	367	13%	21%	857	11%	18%	1,042	12%	18%
R51	2625	-49.20%	72.27%	2268	11%	94%	1761	64%	74%	367	-15%	81%	857	9%	86%	1,042	36%	83%
R52	2625	34.63%	51.35%	2268	32%	43%	1761	27%	41%	367	21%	69%	857	21%	64%	1,042	25%	62%
R53	2625	-12.54%	40.01%	2268	0%	49%	1761	17%	51%	367	-6%	35%	857	1%	41%	1,042	7%	40%
R54	2625	14.30%	25.09%	2268	11%	23%	1761	9%	19%	367	9%	37%	857	10%	33%	1,042	6%	26%
R56	2625	-42.22%	38.52%	2268	-14%	37%	1761	5%	29%	367	-34%	54%	857	-23%	51%	1,042	-10%	48%
R58	2625	-33.34%	73.07%	2268	16%	85%	1761	57%	66%	367	-11%	77%	857	10%	78%	1,042	32%	73%
R60	2625	19.49%	27.44%	2268	19%	26%	1761	19%	27%	367	14%	27%	857	16%	29%	1,042	17%	29%
R61	2625	-28.20%	39.75%	2268	-14%	39%	1761	2%	34%	367	-12%	39%	857	-9%	41%	1,042	2%	41%
R62	2625	50.98%	27.78%	2268	43%	25%	1761	36%	22%	367	50%	36%	857	42%	35%	1,042	36%	33%
R63	2625	86.91%	11.28%	2268	78%	13%	1761	71%	11%	367	80%	23%	857	78%	21%	1,042	74%	21%
R65	2625	49.04%	22.40%	2268	50%	22%	1761	49%	22%	367	47%	28%	857	50%	27%	1,042	51%	26%
R67	2625	8.32%	19.21%	2268	4%	12%	1761	5%	12%	367	19%	37%	857	16%	31%	1,042	16%	28%
R68	2625	51.02%	28.13%	2268	64%	25%	1761	73%	21%	367	57%	34%	857	61%	31%	1,042	69%	27%
R69	2625	36.69%	26.78%	2268	44%	25%	1761	51%	25%	367	45%	35%	857	50%	32%	1,042	57%	30%
R70	2625	63.16%	19.29%	2268	63%	17%	1761	62%	17%	367	64%	21%	857	65%	18%	1,042	64%	19%
R72	2625	4.73%	9.85%	2268	3%	7%	1761	3%	7%	367	17%	31%	857	14%	28%	1,042	12%	22%

R73	2625	7.86%	12.89%	2268	3%	7%	1761	3%	6%	367	19%	29%	857	18%	27%	1,042	14%	23%
R75	2625	18.94%	21.62%	2268	23%	22%	1761	24%	22%	367	14%	24%	857	14%	23%	1,042	16%	22%
R78	2625	53.20%	20.54%	2268	55%	19%	1761	56%	19%	367	63%	20%	857	63%	18%	1,042	64%	16%
R79	2625	31.71%	31.31%	2268	34%	30%	1761	36%	30%	367	35%	39%	857	33%	40%	1,042	40%	39%
R80	2625	-7.58%	28.67%	2268	-2%	27%	1761	3%	24%	367	-7%	51%	857	-8%	43%	1,042	1%	38%
CPI	2625	0.57	0.55	2268	0.65	0.55	1761	0.65	0.55	367	0.55	0.58	857	0.52	0.55	1,042	0.48	0.54
LTI	2625	2.59	1.06	2268	2.62	1.06	1761	2.55	0.95	367	2.26	0.92	857	2.30	0.91	1,042	2.39	0.88
IPI	2625	-1.12	3.66	2268	-0.59	3.18	1761	0.40	2.15	367	-0.97	4.39	857	-0.09	3.26	1,042	0.52	1.88
RTV	2625	-0.84	2.41	2268	-0.60	2.02	1761	-0.11	1.24	367	-1.04	2.72	857	-0.29	2.05	1,042	0.19	1.18
TER	2625	0.02	1.20	2268	0.11	1.18	1761	0.42	1.02	367	0.22	1.21	857	0.41	1.05	1,042	0.55	0.87

Panel B			
Healthy firms			
Variable	N. Obs	mean	std
R2	968,736	16%	26%
R3	968,736	17%	35%
R4	968,736	6%	23%
R5	968,736	10%	28%
R6	968,736	42%	61%
R7	968,736	8%	24%
R8	968,736	4%	21%
R9	968,736	8%	22%
R10	968,736	13%	23%
R11	968,736	23%	26%
R12	968,736	35%	31%
R14	968,736	64%	53%
R15	968,736	26%	26%
R16	968,736	30%	30%
R17	968,736	1%	19%
R18	968,736	2%	17%
R19	968,736	14%	27%
R20	968,736	9%	21%
R21	968,736	3%	20%
R22	968,736	5%	11%
R25	968,736	20%	27%
R26	968,736	42%	37%
R27	968,736	3%	18%
R28	968,736	11%	25%
R29	968,736	3%	20%
R30	968,736	19%	39%
R32	968,736	18%	25%
R33	968,736	49%	35%
R35	968,736	8%	18%
R36	968,736	71%	23%
R37	968,736	58%	24%
R38	968,736	31%	47%
R40	968,736	29%	39%
R41	968,736	20%	36%
R42	968,736	28%	32%
R43	968,736	54%	33%
R46	968,736	11%	22%
R47	968,736	41%	41%
R48	968,736	46%	39%
R49	968,736	18%	19%
R51	968,736	94%	27%
R52	968,736	23%	54%

R53	968,736	12%	25%
R54	968,736	3%	15%
R56	968,736	28%	21%
R58	968,736	52%	46%
R60	968,736	21%	32%
R61	968,736	30%	39%
R62	968,736	21%	20%
R63	968,736	57%	17%
R65	968,736	50%	20%
R67	968,736	25%	30%
R68	968,736	85%	17%
R69	968,736	74%	24%
R70	968,736	65%	16%
R72	968,736	19%	25%
R73	968,736	11%	14%
R75	968,736	16%	19%
R78	968,736	59%	18%
R79	968,736	36%	30%
R80	968,736	9%	18%
CPI	968,736	0.53	0.56
LTI	968,736	2.25	0.91
IPI	968,736	-1.03	4.29
RTV	968,736	-0.92	2.73
TER	968,736	0.21	1.18

Appendix B

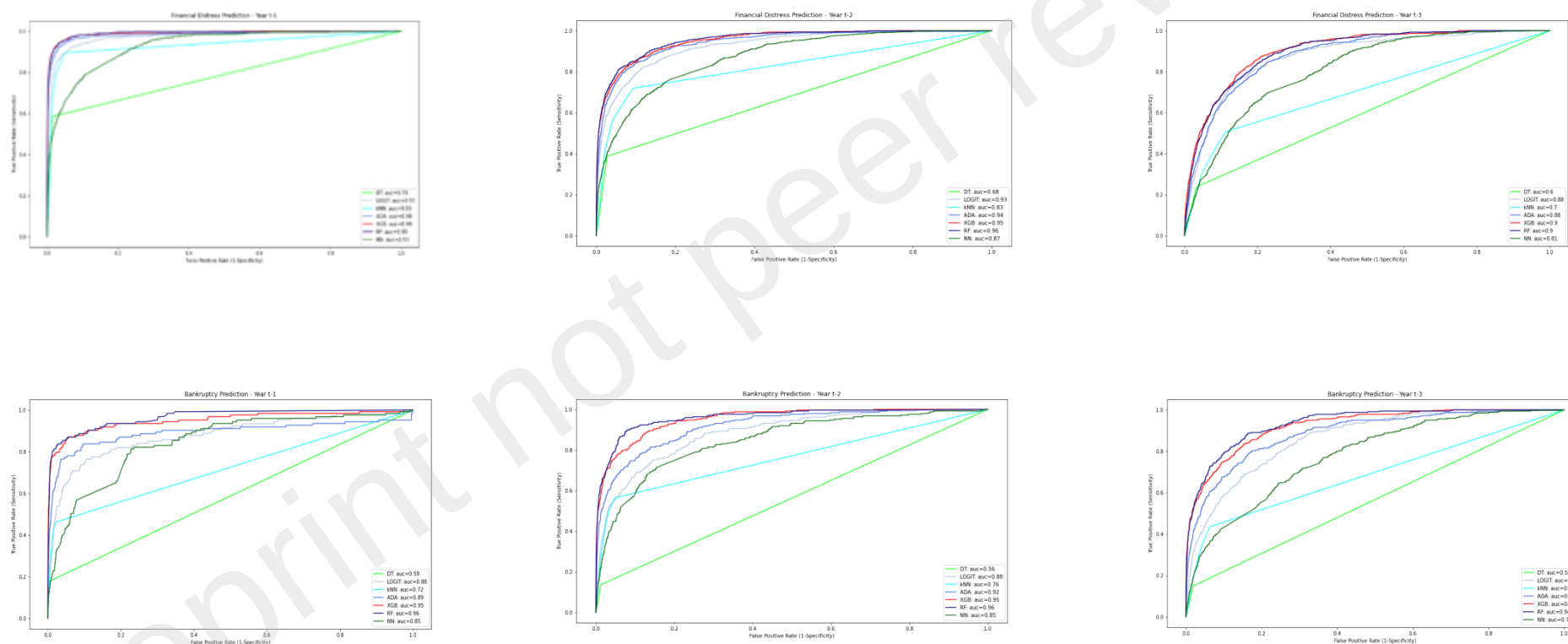


Fig. 1B. ROC curves of all the models.

The figure shows the ROC curves for the financial distress and bankruptcy models at time $t-1$, $t-2$, and $t-3$. The models include DTs, logit, kNN, AdaBoost, XGBoost, RF, and NN.

Table 1B.**Logit model results at time $t-2$ and $t-3$.**

This table shows the estimates for financial distress (FDP) and bankruptcy prediction (BKP) (at $t-2$, and $t-3$). Significance levels: '10%, *5%, **1%, ***0.1%.

Code	Variable	FDP $t-2$	S	FDP $t-3$	S
(Intercept)		-9.141	***	-10.849	***
R2	Cash flow/Short-term debt	-2.183	***		
R3	Interest expense/Cash flow	0.358	***	0.317	***
R4	Cash flow/Sales	-0.255	'	-0.519	***
R5	After-tax return on total equity (ROE)	-0.412	***		
R6	After-tax return on common equity (ROE)	-0.718	***	-1.135	***
R8	Profit margin after depreciation/Sales	-0.594			
R9	Profit margin before depreciation/Sales	0.967	**	0.285	
R10	Return on investment (ROI)	0.340	*		
R11	Service expenses/Sales	-0.596	***	-0.773	***
R12	Raw material expenses/Sales	0.821	***	0.869	***
R14	Labor expenses/EBITDA	-0.317	***	-0.327	***
R15	Added value/Sales				
R16	Effective tax rate	-0.146		0.128	
R18	Third use assets/Sales			-0.998	**
R19	Return on capital (invested)	-0.208	'		
R21	Pre-tax profit margin				
R22	Return on assets	-0.932	*		
R25	EBITDA/Current liabilities	-0.786	'	-1.828	***
R26	Net income/EBIT	0.134	'		
R27	Amortization and depreciation/Sales			0.721	**
R28	Cash and cash equivalents/Sales	-1.098	***	-0.765	***
R29	Fixed assets/Sales (long-term assets turnover)				
R30	Net working capital/Sales	-0.181			
R32	Inventory over sales (with value of production)			-0.553	*
R33	Payables turnover	0.474	***	0.334	**
R35	Stockholders equity/Sales			0.592	**
R36	Total assets/Sales (with value of production)	2.059	***	2.598	***
R37	Current assets/Sales (Current assets turnover)	-1.562	***	-1.581	***
R38	Quick/Liquid assets over sales	-0.198	**	-0.230	**
R40	Inventory over sales (inv. Inventory Turnover) (var)	0.404	***	0.750	***
R41	Receivables over sales (inv. Receivable Turnover) (var)	0.434	***	0.491	***
R42	Payables over expenditures (inv. Payables Turnover) (var)	0.812	***	0.629	***
R43	Total liabilities/Sales	1.315	***	1.235	***
R46	Common equity/Invested capital	0.368	**	0.358	*
R47	Total debt/Invested capital			0.316	**
R48	Fixed assets/Total equity	0.366	**	0.438	***
R49	Fixed assets/Total assets	-1.103	***	-2.159	***
R51	Total assets/Total equity (Net worth)	-1.107	***	-0.492	***
R52	Interest expense/Long-term debt (cost of debt)				
R53	Interest expense/EBITDA	0.489	***	0.799	***

R54	Interest expense/Sales	0.447	'	-0.436	'
R58	Debt/Equity	0.187	'		
R60	Total debt/Common equity	0.743	***	0.544	***
R61	Total debt/EBITDA				
R62	Total debt/Total assets	3.197	***	2.766	***
R63	Total liabilities/Total assets	5.725	***	5.149	***
R65	Receivables/Current assets	0.990	*	1.814	***
R67	Cash ratio	-1.417	**	-1.349	*
R68	Current ratio	2.106	***	3.108	***
R69	Quick ratio (acid test)	-1.076	**	-1.642	***
R70	Current indebtedness ratio	1.961	***	1.531	***
R72	Cash balance/Total liabilities	1.447		2.048	*
R73	Cash and cash equivalents/Total assets	-4.258	***	-3.876	***
R78	Current assets/Total assets	-1.725	**	-1.830	**
R80	Added value per employee	0.103	***	0.059	'
CPI	Consumer prices: all items	-0.202	*	-0.714	***
LTI	Long-term interest rate	0.292	***	1.050	***
IPI	Industrial production, s.a.	0.108	***	0.391	***
RTV	Retail trade (Volume), s.a.	0.096	**	0.105	**
TER	Total employment: all persons, s.a.	-0.317	**		
AGR	Agriculture & Mining, Quarrying, and Oil and Gas Extraction	-1.368	***	-1.004	***
UTI	Utilities	0.598	**		
CON	Construction	0.263	*		
MAN	Manufacturing	0.390	***	0.242	***
TRA	Wholesale + Retail Trade	0.154			
TRS	Transportation & Warehousing				
NE	Northeast macro-region				
C	Center macro-region				
SI	South-Islands macro-region	-0.148	'		

Table 2B.**Logit model results at time $t-2$ and $t-3$.**

This table shows the estimates for financial distress (FDP) and bankruptcy prediction (BKP) (at $t-2$, and $t-3$). Significance levels: '10%, *5%, **1%, ***0.1%.

Ratio	Variable	FDP $t-2$	S	FDP $t-3$	S
(Intercept)		-1.192		-0.109	
R2	Cash flow/Short-term debt	1.662	***	1.298	**
R3	Interest expense /Cash flow	0.649	***	0.158	
R4	Cash flow/Sales	-0.847	***	-0.675	***
R5	After-tax return on total equity (ROE)	-1.147	***	-0.638	***
R6	After-tax return on common equity (ROE)	-0.816	***	-0.287	**
R7	EBITDA/Sales	-0.891	***	-0.356	'
R8	Profit margin after depreciation/Sales	2.164	***	1.129	*
R9	Profit margin before depreciation/Sales	-2.545	***	-2.431	***
R10	Return on investment (ROI)	0.417	*		
R11	Service expenses/Sales	-0.587	*	-0.816	***
R14	Labor expenses/EBITDA	-1.041	***	-0.927	***
R15	Added value/Sales	0.823	***	0.704	***
R16	Effective tax rate	-0.416	**	-0.176	
R17	Net profit margin	1.098			
R18	Third use assets/Sales	-1.026	**		
R19	Return on capital (invested)				
R20	Pre-tax return on net operating assets	0.288			
R21	Pre-tax profit margin	-0.986		-0.344	
R22	Return on assets			1.353	**
R25	EBITDA/Current liabilities	2.256	***	1.376	**
R26	Net income/EBIT	0.188		-0.175	
R28	Cash and cash equivalents/Sales	-1.476	***		
R29	Fixed assets/Sales (long-term assets turnover)				
R30	Net working capital/Sales	-0.283	'	-0.579	***
R33	Payables turnover			0.338	*
R35	Stockholders equity/Sales			0.348	
R36	Total assets/Sales (with value of production)			-0.653	**
R37	Current assets/Sales (Current assets turnover)	-1.283	***		
R38	Quick/Liquid assets over sales	0.166			
R40	Inventory over sales (inv. inventory turnover) (var)			0.232	*
R41	Receivables over sales (inv. receivable turnover) (var)	0.795	***	0.306	**
R42	Payables over expenditures (inv. payables turnover) (var)	1.063	***	0.346	**
R43	Total liabilities/Sales	1.670	***	0.555	**
R46	Common equity/Invested capital	0.798	***		
R47	Total debt/Invested capital				
R48	Fixed assets/Total equity			0.405	**
R49	Fixed assets/Total assets			-1.183	**
R51	Total assets/Total equity (Net worth)	-1.112	***	-1.555	***
R52	Interest expense/Long-term debt (cost of debt)				
R53	Interest expense/EBITDA	1.108	***	0.956	***

R54	Interest expense/Sales			-1.113	***
R56	Total equity/Total assets	-2.723	***	-2.494	***
R58	Debt/Equity			0.452	***
R60	Total debt/Common equity				
R61	Total debt/EBITDA	-0.595	***	-0.464	***
R62	Total debt/Total assets	1.036	***		
R63	Total liabilities/Total assets	-3.323	***	-3.741	***
R65	Receivables/Current assets	6.840	***	4.455	***
R67	Cash ratio			-2.237	***
R68	Current ratio	1.199	'	1.366	*
R69	Quick ratio (acid test)	-3.779	***	-3.209	***
R70	Current indebtedness ratio	-1.270	**	-2.646	***
R72	Cash balance/Total liabilities				
R73	Cash and cash equivalents/Total assets	6.285	***	4.955	***
R75	Inventories/Total assets	1.710	*		
R78	Current assets/Total assets	3.241	***	2.885	***
R79	Sales per employee	-0.012		0.010	
R80	Added value per employee			0.136	**
CPI	Consumer prices: all items	-1.282	***	-2.072	***
LTI	Long-term interest rate			1.499	***
IPI	Industrial production	0.569	***	0.643	***
RTV	Retail trade (Volume)			0.104	*
TER	Total employment: all persons	-1.499	***	-0.496	***
AGR	Agriculture & Mining, Quarrying, and Oil and Gas Extraction	0.738	**	0.610	**
CON	Construction	0.674	***	0.386	***
MAN	Manufacturing	-0.244	'	-0.217	*
UTI	Utilities	-0.902	*	-0.768	**
TRA	Wholesale + retail trade	0.370	*		
TRS	Transportation & warehousing				
NE	Northeast macro-region	0.313	**	-0.295	**
C	Center macro-region	0.164			
SI	South-Islands macro-region				

Table 3B.

PFI scores and confidence intervals according to Random Forest.

FDP (<i>t-1</i>)		FDP (<i>t-2</i>)		FDP (<i>t-3</i>)	
Weight	Feature	Weight	Feature	Weight	Feature
0.0087 ± 0.0002	R48	0.0053 ± 0.0002	R3	0.0032 ± 0.0001	RTV
0.0082 ± 0.0002	R51	0.0039 ± 0.0001	R5	0.0018 ± 0.0001	IPI
0.0081 ± 0.0004	R56	0.0031 ± 0.0003	R17	0.0016 ± 0.0001	R3
0.0048 ± 0.0004	R63	0.0022 ± 0.0001	R6	0.0014 ± 0.0001	R6
0.0012 ± 0.0001	R5	0.0016 ± 0.0000	R51	0.0009 ± 0.0001	CPI
0.0003 ± 0.0000	R14	0.0011 ± 0.0002	R14	0.0009 ± 0.0001	TER
0.0003 ± 0.0000	R19	0.0006 ± 0.0001	RTV	0.0009 ± 0.0000	LTi
0.0002 ± 0.0000	RTV	0.0004 ± 0.0000	R26	0.0008 ± 0.0000	R5
0.0001 ± 0.0000	TER	0.0004 ± 0.0000	R61	0.0005 ± 0.0001	R17
0.0001 ± 0.0000	R3	0.0003 ± 0.0000	LTi	0.0002 ± 0.0000	R26
0.0001 ± 0.0000	R10	0.0003 ± 0.0000	R20	0.0002 ± 0.0000	R20
0.0001 ± 0.0000	R61	0.0003 ± 0.0000	R19	0.0001 ± 0.0001	R53
0.0001 ± 0.0000	R20	0.0002 ± 0.0000	R53	0.0001 ± 0.0000	R16
0.0001 ± 0.0000	R25	0.0002 ± 0.0001	IPI	0.0001 ± 0.0000	R14
0.0001 ± 0.0000	R53	0.0002 ± 0.0001	CPI	0.0001 ± 0.0000	R19
0.0001 ± 0.0001	CPI	0.0002 ± 0.0000	R16	0.0001 ± 0.0000	MAN
0.0001 ± 0.0001	R41	0.0002 ± 0.0000	TER	0.0001 ± 0.0000	R51
0.0001 ± 0.0001	IPI	0.0001 ± 0.0000	R48	0.0000 ± 0.0000	C
0.0000 ± 0.0000	R29	0.0001 ± 0.0000	R10	0.0000 ± 0.0000	R48
0.0000 ± 0.0000	R58	0.0001 ± 0.0001	MAN	0.0000 ± 0.0000	R61

BKP (<i>t-1</i>)		BKP (<i>t-2</i>)		BKP (<i>t-3</i>)	
Weight	Feature	Weight	Feature	Weight	Feature
0.0017 ± 0.0000	R51	0.0112 ± 0.0002	R3	0.0049 ± 0.0001	RTV
0.0006 ± 0.0001	R56	0.0102 ± 0.0002	R51	0.0015 ± 0.0000	R51
0.0005 ± 0.0000	R48	0.0084 ± 0.0002	R5	0.0007 ± 0.0000	R5
0.0002 ± 0.0000	R5	0.0026 ± 0.0000	R6	0.0005 ± 0.0001	CPI
0.0001 ± 0.0000	C	0.0025 ± 0.0002	R17	0.0003 ± 0.0001	LTi
0.0001 ± 0.0000	SI	0.0009 ± 0.0000	R56	0.0002 ± 0.0000	R56
0.0001 ± 0.0000	R10	0.0003 ± 0.0000	R48	0.0002 ± 0.0000	R3

0.0001 ± 0.0000	R20
0.0000 ± 0.0000	R19
0.0000 ± 0.0000	R2
0.0000 ± 0.0000	R58
0.0000 ± 0.0000	RTV
0.0000 ± 0.0000	CON
0.0000 ± 0.0000	R46
0.0000 ± 0.0000	R26
0.0000 ± 0.0000	R53
0.0000 ± 0.0000	R32
0.0000 ± 0.0000	R38
0.0000 ± 0.0000	MAN
0.0000 ± 0.0000	R7

0.0002 ± 0.0000	R2
0.0002 ± 0.0000	R20
0.0002 ± 0.0000	C
0.0001 ± 0.0000	R14
0.0001 ± 0.0000	SI
0.0001 ± 0.0000	R19
0.0001 ± 0.0000	R72
0.0001 ± 0.0000	R58
0.0001 ± 0.0000	R70
0.0001 ± 0.0000	R25
0.0000 ± 0.0000	R10
0.0000 ± 0.0000	R46
0.0000 ± 0.0000	R62

0.0002 ± 0.0000	R48
0.0001 ± 0.0000	C
0.0001 ± 0.0000	R58
0.0001 ± 0.0000	R20
0.0001 ± 0.0000	SI
0.0001 ± 0.0000	R6
0.0000 ± 0.0000	R52
0.0000 ± 0.0000	R73
0.0000 ± 0.0000	R46
0.0000 ± 0.0000	R18
0.0000 ± 0.0000	R19
0.0000 ± 0.0000	TRA
0.0000 ± 0.0000	R53

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